

Pathology of Cerebrovascular Diseases



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

William Zaloga, D.O.

Assistant Professor

Department of Clinical Medicine

New York Institute of Technology

at Arkansas State University

Wilson Hall, Room 424F

Office #: (870) 680-8899

wzaloga@nyit.edu

Session Objectives

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

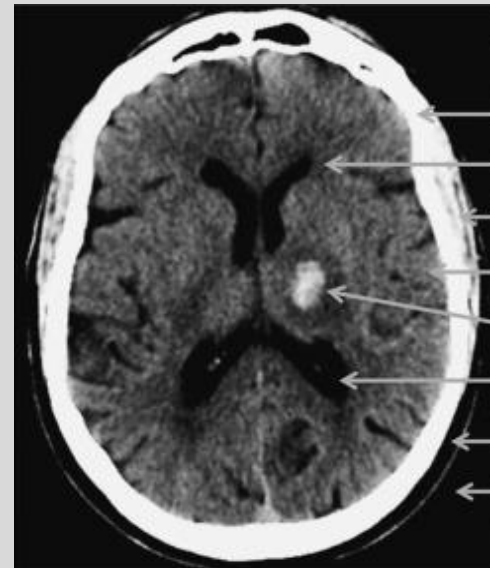
1. Describe **cerebrovascular pathology**, including **ischemic stroke**, **hemorrhagic stroke**, **transient ischemic attack**, and **brain trauma**.
2. Outline the **histological changes in ischemic infarction** based on the time since the ischemic event.
3. Explain the **etiologies and classifications of ischemic strokes**, including **thrombotic**, **embolic**, and **global hypoperfusion**.
4. Classify and differentiate **hemorrhagic strokes**:
 1. Intracerebral hemorrhage (ICH): **Hypertension vs. Cerebral Amyloid Angiopathy (CAA)**.
 2. Subarachnoid hemorrhage (SAH): **Traumatic, Aneurysmal, and Arteriovenous Malformations (AVMs)**.
5. Identify and describe the pathology of cerebrovascular trauma, including **brain herniation**, **skull fractures**, **concussion**, **subdural**, and **epidural hematomas**.
6. Discuss and describe the pathogenesis and morphological principles of direct parenchymal injuries, including **contusion**, **coup-contrecoup injury**, **diffuse axonal injury**.



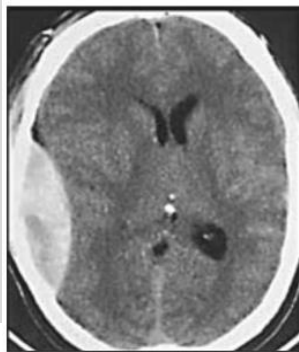
HEAD CT (w/o contrast) quick reference

NOT ON TEST – good to know

- HU: Measures tissue density, **water = 0 HU**, **air = -1000 HU**
- Less dense ones like **fat appearing dark (negative HU)**
- Denser materials like **bone appear bright (high positive HU)**
- **Blood has high HU** due to high protein and iron content



CT Number Ranges in Hounsfield Units (HU)	
Bone	+1000
White matter	+20 to 30
Muscle	+20 to 40
Gray matter	+30 to 40
Hemorrhage	+65 to +95
CSF (water)	0
Fat	-30 to -70
Air	-1000



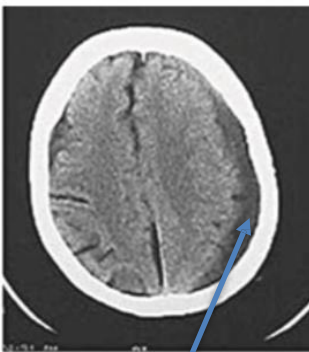
Acute

HYPERDENSE



Subacute

ISODENSE



Chronic

HYPODENSE

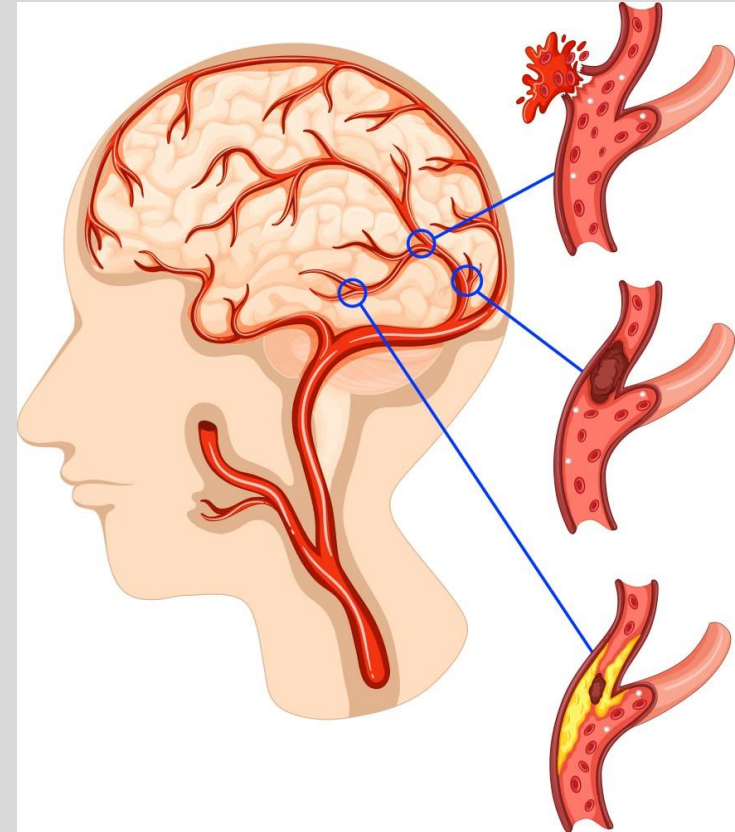


<https://ddxof.com/ct-interpretation-head/>

Cerebrovascular disease (CVD)

- Common in US
- **All disorders** with an area of the **brain temporarily or permanently affected by ischemia or bleeding** and one or more **cerebral blood vessels are involved** in the pathological processes
- **AKA Stroke** = Brain blood **flow** suddenly reduced with cell hypoxia due to **hypoxemia** (low blood O₂) or **ischemia** (low blood flow) causes cell disability and **death/infarction**:
- When blood to the brain is reduced, tissue **survival depends** on **collateral** circulation presence, ischemia **duration** (sustained → infarction) & **magnitude** and **rapidity** of reduction
- Collateral flow source is **circle of Willis**, external carotid-ophthalmic artery collaterals - **Little if any** collateral flow for thalamus, basal ganglia & deep white matter deep penetrating vessels

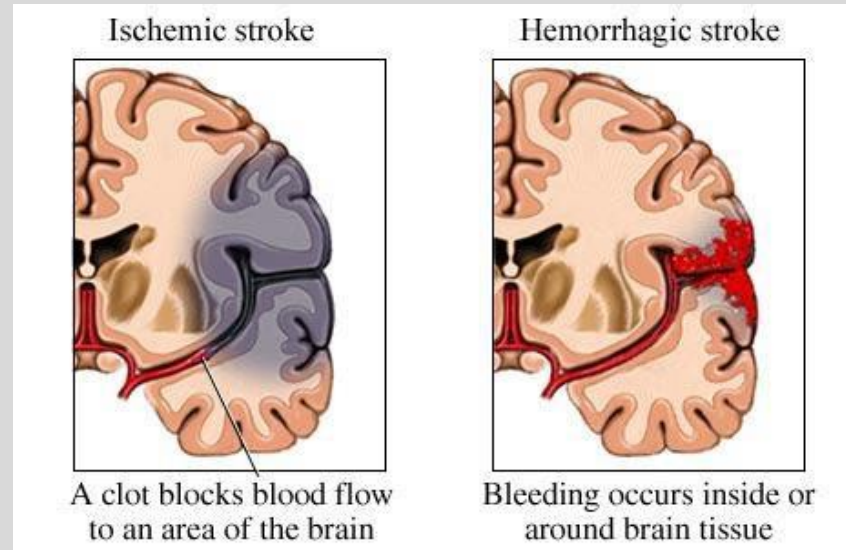
NEW YORK INSTITUTE
OF TECHNOLOGY



Different types of strokes/ cerebrovascular accidents (**CVA**)

Ischemic stroke

- **87%**
- Caused by **blockage of an artery** (rare cases vein), result = lack of perfusion
- **Hemorrhagic stroke**
 - **13%**
 - Caused by **bleeding** cerebral artery



<https://figures.semanticscholar.org/a14f6c3aef5265960e7fd1f29ecbbe3cc7d72986/23-Figure1.1-1.png>

- The **transition region** between necrotic tissue and normal brain is an area of “**at-risk**” **brain**, referred to as the ***penumbra*** - **can be rescued** from cell death with anti-apoptotic interventions, suggesting ischemic neurons die by apoptosis and necrosis

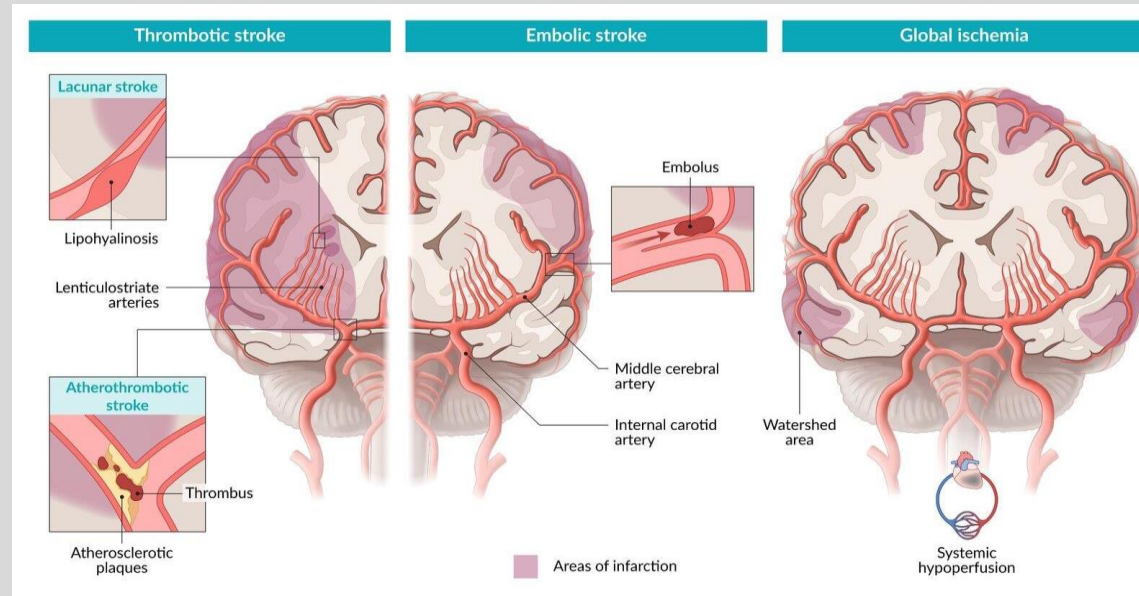
Ischemic stroke etiology

- Arterio-/arteriolosclerosis, atherosclerosis may produce carotid/ vertebral/ and intracranial stenosis (stenotic/ turbulent areas at risk):
- **Thrombotic: Atherosclerosis** (vulnerable plaque acute in-situ thrombus like CAD) & **Lipohyalinosis** (associated with **hypertension**)
- **Embolic: Cardiogenic** mostly (ex. MI, valve dz, afib. w/ cardiac mural thrombi) vs **non-cardiogenic** (ex. usually from atherosclerotic plaques in internal carotid artery)
- **Middle cerebral** artery territory most frequently affected by **emboli** which tend to lodge at vessel branch or in pre-existing luminal **stenosis**
- Other less common causes - pathology discussed in another class:
- Vasculitides may also cause luminal narrowing & occlusion (ex. Polyarteritis nodosa)
- Drug abuse may cause vascular spasms and occlusion
- Aneurysms, vascular malformations, coagulopathies

- **Global ischemia:** Watershed zones (border zones) are at junctions of two major artery territories (vulnerable to ischemia) – there's a **later slide** about this

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

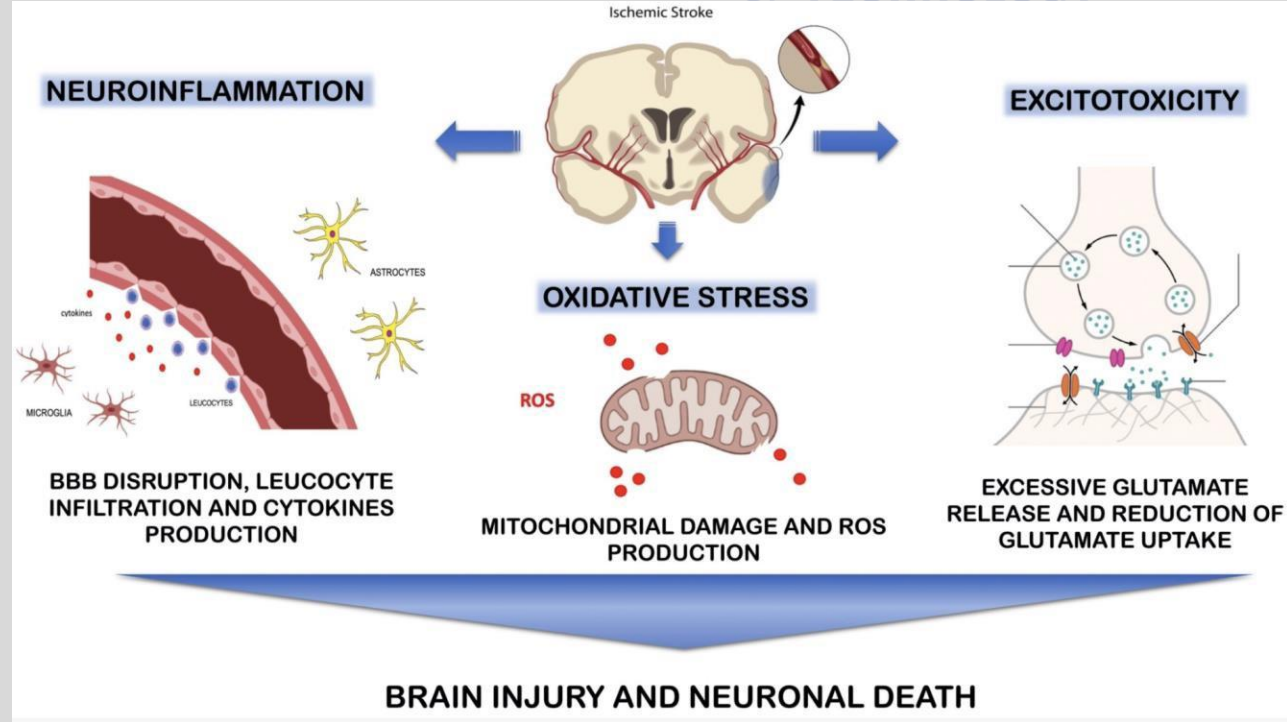


<https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.amboss.com%2Fus%2Fknowledge%2Foverview-of-stroke&psig=AOvVaw2J9dN5xyP7kuQEHQBkVLG3&ust=1768134836160000&source=images&cd=vfe&opi=89978449&ed=0CBIOjRxqFwoTCMC-qNv9gIJDFOAAAAAdAAAAABAE>

Ischemic stroke molecular pathogenesis

NEW YORK INSTITUTE
OF TECHNOLOGY

- Ischemia can cause **excitatory amino acid** neurotransmitters inappropriate release such as **glutamate** - can **damage neurons** by **allowing excessive influx of calcium ions** through N-methyl-D- aspartate (NMDA)-type glutamate receptors (phenomenon **called excitotoxicity**)



DO NOT MEMORIZE

Maida CD, Norrito RL, Rizzica S, Mazzola M, Scarantino ER, Tuttolomondo A. Molecular Pathogenesis of Ischemic and Hemorrhagic Strokes: Background and Therapeutic Approaches. Int J Mol Sci. 2024 Jun 7;25(12):6297. doi: 10.3390/ijms25126297. PMID: 38928006; PMCID: PMC11203482

Ischemic stroke

- Acute neurological condition with arterial supply is impaired - cause is **occlusion of an upstream artery** → possible **infarction** (pathological process results in necrotic tissue due to acute interruption of blood flow)
- Ischemia → tissue **hypoxia** → **irreversible** neuronal injury (**can be as fast** as 5 min after hypoxia)
- Sustained ischemia → **infarction** → **liquefactive necrosis** (digestion of dead cells by hydrolytic enzymes from inflammatory cells or microorganisms with complete loss of cellular architecture, results in liquid viscous mass)
- Over time, **macrophages** clear the debris, leading to the formation of a **cystic space**
- Grossly, little change in first 6 hours of irreversible injury: 48 hours, pale, soft, and swollen, gray-white matter junction indistinct; 2-10 days, brain becomes gelatinous and friable; 10 days-3 weeks, tissue liquefies, leaving fluid-filled (reabsorbed) cavity

<https://en.wikipedia.org/wiki/Stroke>

NEW YORK INSTITUTE
OF TECHNOLOGY



CT Head

Ischemic stroke molecular events

NEW YORK INSTITUTE
OF TECHNOLOGY

pathic

TIME SINCE ISCHEMIC

EVENT

12–24 HOURS

24–72 HOURS

3–5 DAYS

1–2 WEEKS

> 2 WEEKS

Histologic
features

Eosinophilic
cytoplasm
+ pyknotic
nuclei (red
neurons)

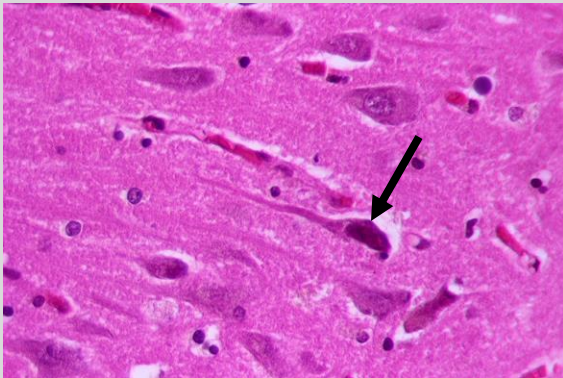
Necrosis +
neutrophils

Macrophages
(microglia)

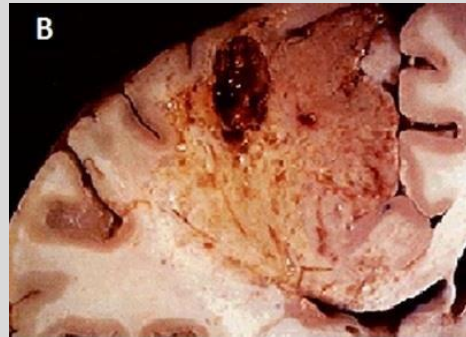
Reactive gliosis
(astrocytes)
+ vascular
proliferation

Glial scar

Cerebral cortex shows neuron loss
and reactive astrocytes, microglia,
fibroblasts, endothelial cells

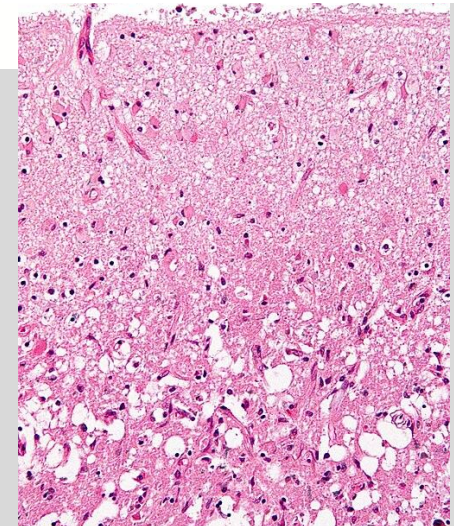


https://en.wikipedia.org/wiki/File:Hypoxic_neuronal_injury,_HE_1.jpg



Liquefactive necrosis in brain tissue

<https://www.alamy.com/stock-image-liquefactive-necrosis-in-brain-tissue-165987706.html>

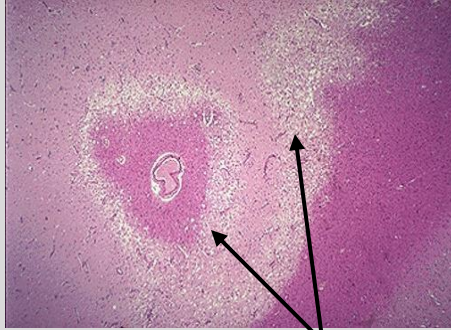


https://en.wikipedia.org/wiki/Glial_scar

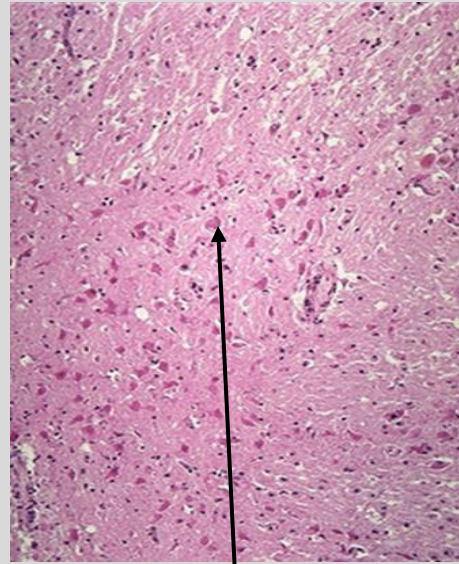
Ischemic stroke histological changes

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

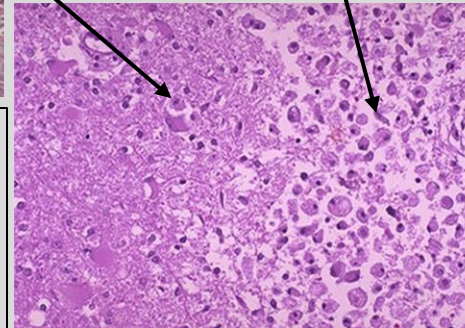


- Acute cerebral infarction: **Marked edema (pale areas)**
- 6-24 hours, eosinophilic neuronal necrosis (↑ cytoplasm **eosinophilia** & nuclear pyknosis/ karyorrhexis - “**dead red neurons**”)
- Reactive **gliosis** begins and recedes after several months: Glial cells, mainly astrocytes, swell, and myelinated fibers begin to disintegrate

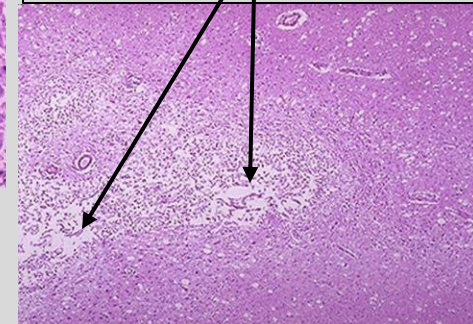


- Anoxic injury: **Red neurons** dying as a result of hypoxia
- ≤48 hours, **neutrophilic** emigration progressively increases, then falls off

- >48 hours to 2-3 wks., macrophages & activated microglia predominate
- Foamy **macrophages** cleaning up lipid debris from the **liquefactive necrosis**



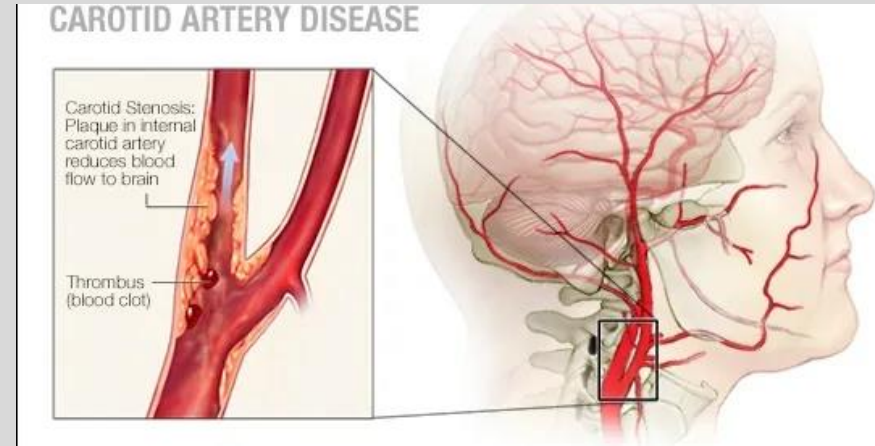
- Surrounded by **residual gliosis** is **resolution** of **liquefactive** necrosis that leaves **cystic spaces**



1. Red Neurons
2. Neutrophils
3. Macrophages
4. Liquefactive necrosis
5. Cystic cavity

Thrombotic ischemic stroke: Atherosclerotic cerebrovascular disease

- Most common type of stroke (~40%)
- Associated with **hypertension** and **diabetes**
- Caused by **thrombus** (blood clot that may fragment and embolize) developing in **arteries supplying the brain** that cause **progressive luminal narrowing**
- Seen in **older persons**, especially those with **high cholesterol and atherosclerosis**
- Common sequence of events: Atherosclerotic plaque rupture → thrombus formation → ischemia → infarct
- Most common at carotid bifurcation, middle cerebral artery origin & either end of the basilar artery

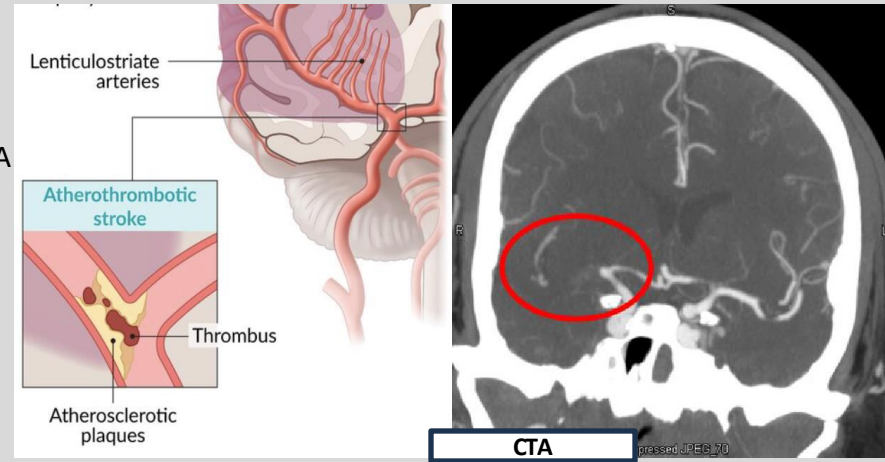
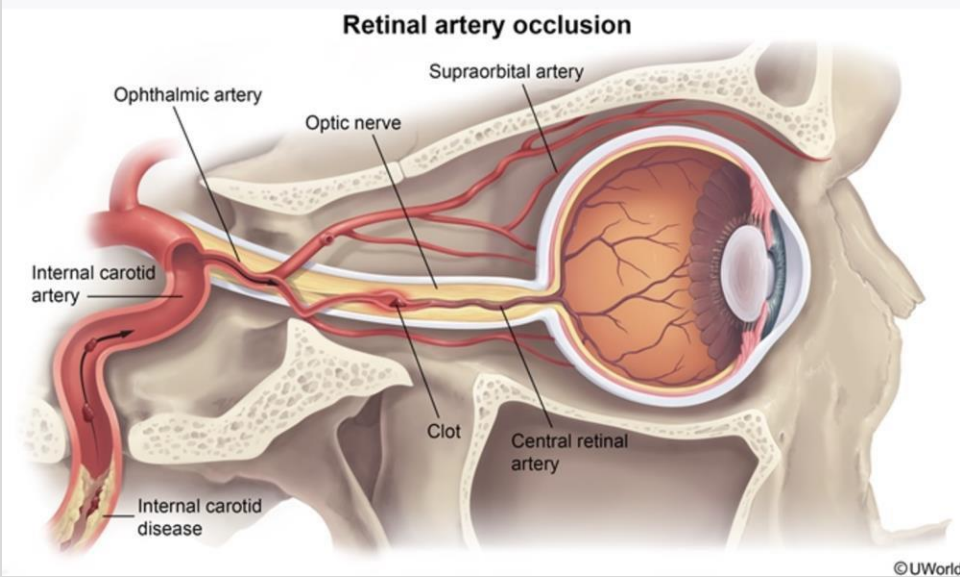


<https://interventionalneurology.com/carotid-artery-disease>

Thrombotic ischemic stroke: Atherosclerosis

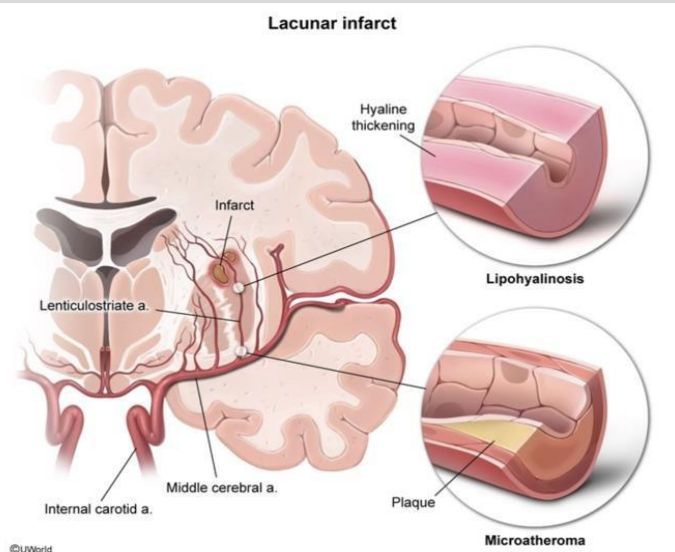
Common sites of occlusion

- **Common Carotid bifurcation**
 - Clot travel up to 1st intracranial branch of ICA, ophthalmic artery → **amaurosis fugax** (transient, seconds-minutes for small emboli/fibrinolysis, vision loss from hypoperfusion of retina or optic nerve)
 - **Middle cerebral artery** - distal to carotid terminus
-
- <https://www.cns.org/nexus/spine/case/acute-ischemic-stroke-from-mca-occlusion>
 - **CT angiogram** of brain with atherothrombotic stroke in the lenticulostriate arteries - red circle, highlights affected **lenticulostriate arteries**
 - Lenticulostriate arteries: Small perforating artery branches from MCA to deep brain structures (basal ganglia (putamen, globus pallidus, caudate nucleus, internal capsule) – common site of small stroke (**lacunar infarcts**) & hemorrhages due to small size and lack of collateral flow → pure **motor or sensory loss**)



Thrombotic ischemic stroke: Lipohyalinosis

- Pathological process affects **small arteries** (arteriosclerosis) and **arterioles** (arteriolosclerosis), associated with **hypertension & diabetes**, can lead to **lacunar strokes**
- Arteries thicken/ narrow due to **fatty (lipo)** and **glassy/proteinaceous (hyaline)** material: Ex. Lenticulostriate arteries accumulation of lipid and hyaline material have thickened arterial walls → narrowing and potential occlusion
- **Chronic hypertensive vasculopathy** → lipohyalinosis of the small vessels → occlusion of small, penetrating arteries → lacunar stroke (**basal ganglia & subcortical white matter most prominent**)



Robbins and Cotran
Pathologic Basis of
Disease, 10th ed. fig.
28.16 *Lacunar
infarcts* in the
caudate and
putamen (*arrows*)

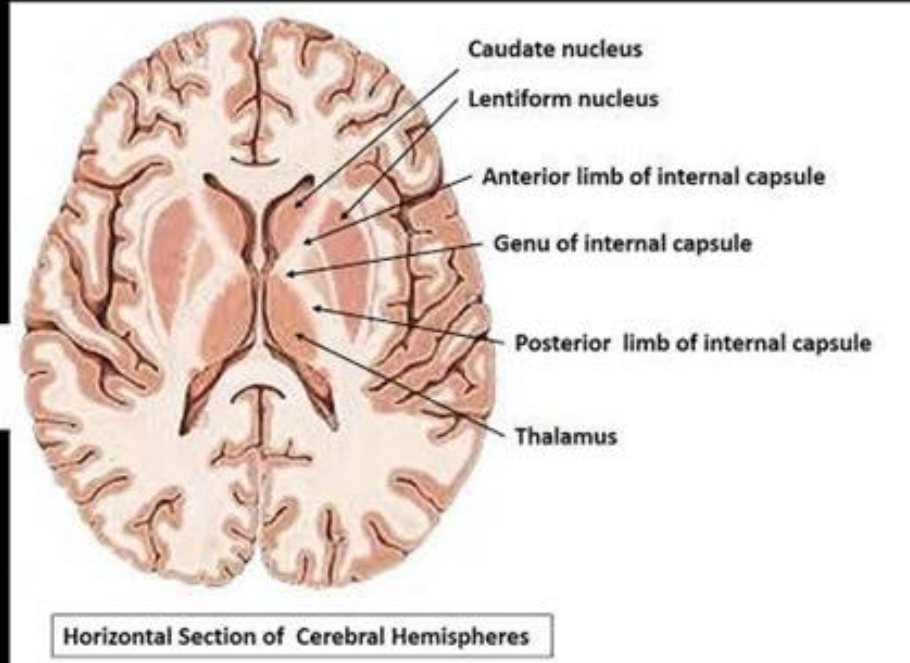
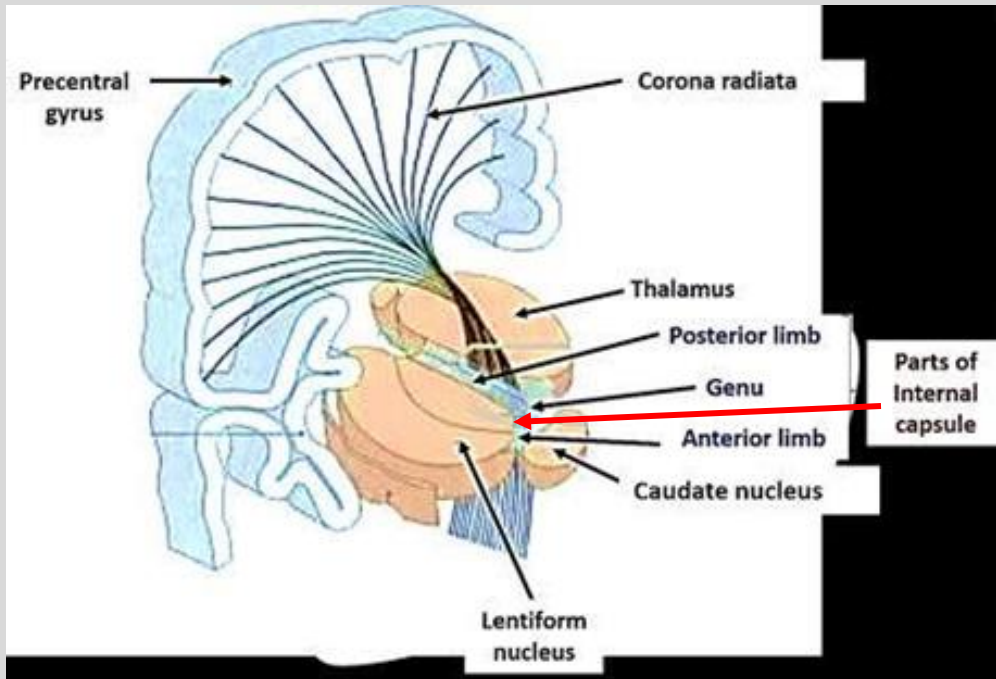


- Named for the characteristic “**lake-like**” appearance of the resulting cavities in the brain tissue on MRI of the brain
- Axial *T2-weighted brain MRI* showing hyperintense signals in the basal ganglia/internal capsule region and other deep brain structures (**hyperintensity causes include stroke**, demyelination, inflammation, or tumors)
- Commonly **affected regions**: Internal capsule, basal ganglia, pons, corona radiata
- Depending on location, clinically **silent or severe impairment**

Internal capsule

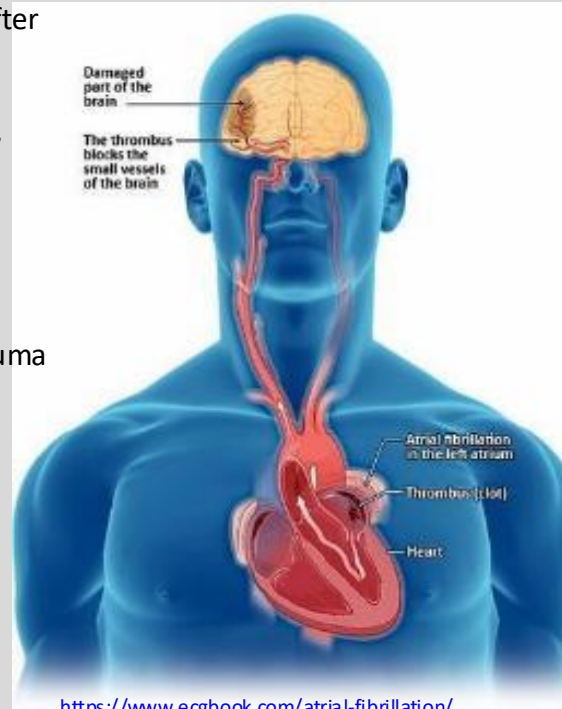
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Embolic ischemic stroke

- ~20% of strokes
- Caused by occlusion of flow due to an **embolus from another part of the body**
- Can affect multiple vascular territories: **Embolic shower** (cardiogenic) - often refers to fat emboli, typically after long bone fracture or orthopedic surgery
 - Fat emboli syndrome Sx/Sx: Blockages in small vessels, mostly in lungs, brain, and skin w/ general cerebral dysfunction (higher cortical function & consciousness, often w/o localizing signs), characteristic white matter hemorrhagic lesions of **bone marrow** emboli after trauma
- **Cardiogenic** causes include:
 - **Atrial fibrillation:** Stasis of blood--> clots
 - Post-MI
 - Mitral valve prolapse
 - Prosthetic valve
 - Atrial myxoma
 - Endocarditis: Septic emboli
 - Rheumatic heart disease

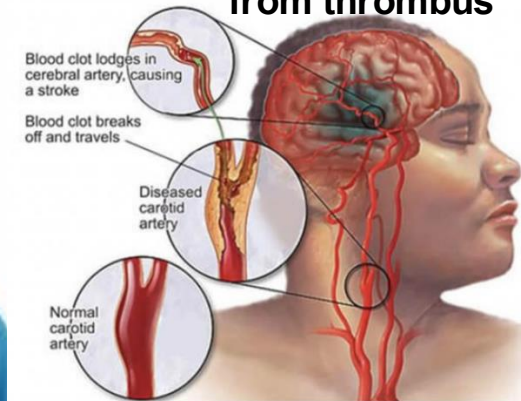


<https://www.ecgbook.com/atrial-fibrillation/>

NEW YORK INSTITUTE
OF TECHNOLOGY

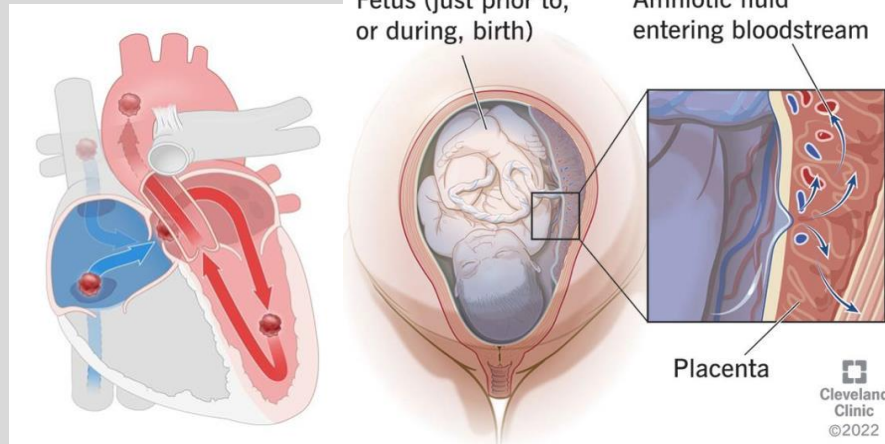
College of Osteopathic
Medicine

Noncardiogenic from thrombus



<https://interventionalneurology.com/carotid-artery-disease>

Embolic ischemic stroke



NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

<https://emedicine.medscape.com/article/460607-overview#a4>

Paradoxical cerebral embolus: If arteriovenous shunt present, ex. heart (**septal**) **defect** present:

- Venous thromboembolism (ex. **DVT**) can become systemic → cardiac septal defect → carotid circulation

Non-cardiogenic/ non-thrombotic cerebral/ pulmonary emboli examples:

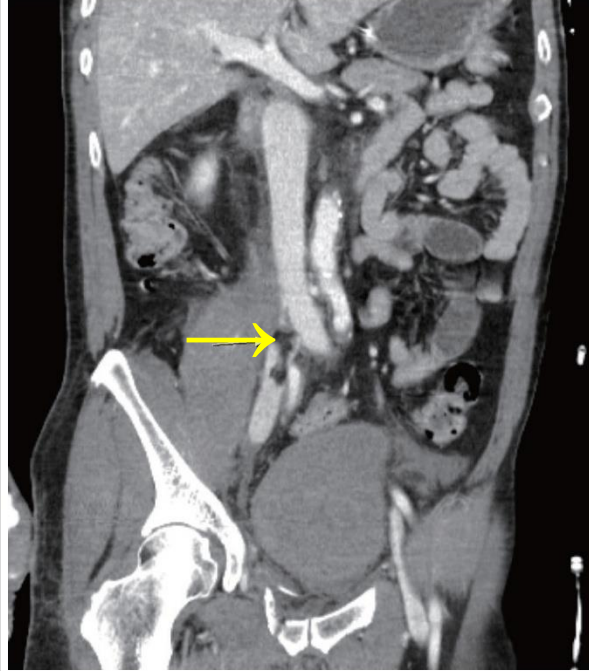
- **Fat embolism:** Globule enter arteries after long bone fracture (in pulmonary capillaries, microscopic globules **deform**, pass to arterial (systemic) circulation reform) - also can bypass lungs through arteriovenous shunts
- **Amniotic fluid embolism:** Amniotic fluid & debris enters mother's prenatal/perinatal circulation (emergency)
- Clinical triad: Respiratory failure (**PE** occurs), focal **neuro.** deficits (due to **anaphylactoid**-like reaction) → pulmonary vasoconstriction, acute right heart failure, hypotension, and hypoxemia) & petechial rash (**DIC** and vascular spasm → focal cerebral ischemia (stroke) or hemorrhage)
- Also, tumor and air (ex. via syringe) have produced emboli

Embolic ischemic stroke - intravenous fat embolism

- Rarely seen on CT:

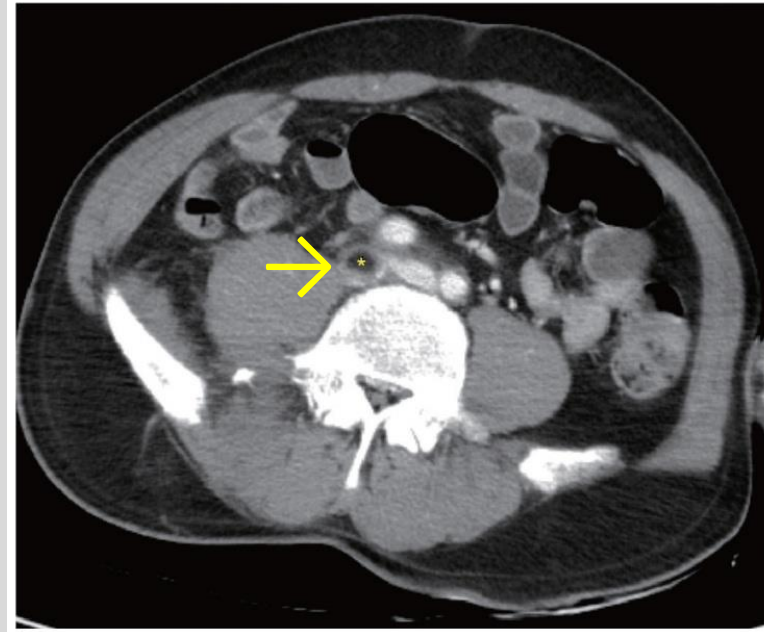


<https://www.jtraumainj.org/upload//thumbnails/jti-31-103f2.jpg> 3D CT reconstruction of femur shaft fracture - longest and strongest bone in the body, significant force required to break it (ex. motor vehicle collision)



<https://www.jtraumainj.org/upload//thumbnails/jti-31-103f5.jpg> Fat embolism in his right common iliac vein was shown on the coronal plane of abdominal computed tomography (arrow)

NEW YORK INSTITUTE
OF TECHNOLOGY

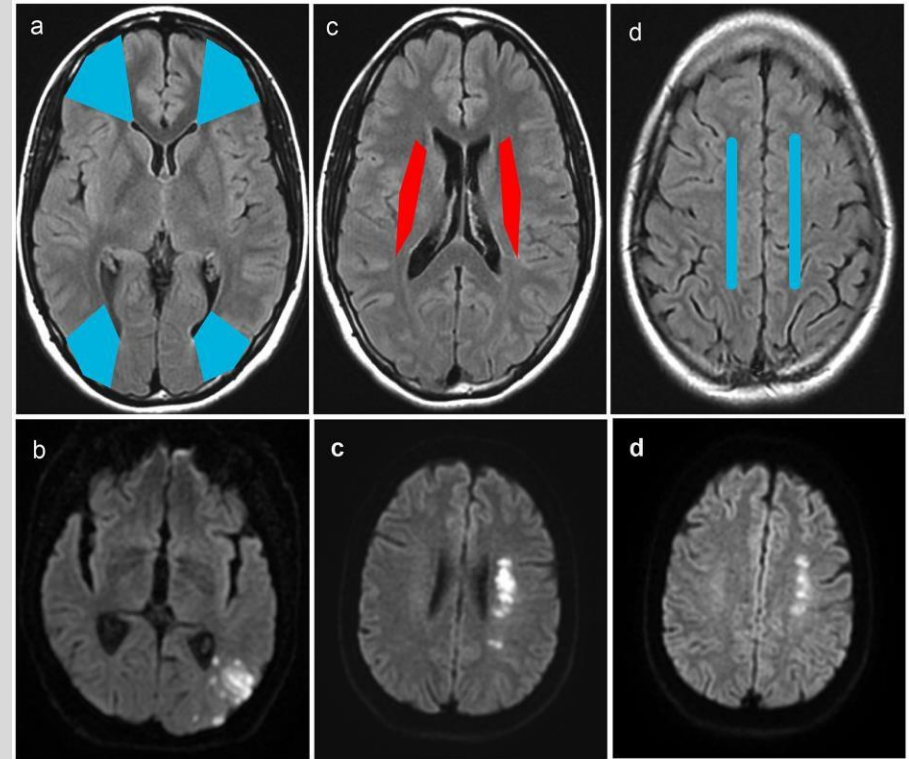


<https://www.jtraumainj.org/upload//thumbnails/jti-31-103f4.jpg> Fat embolism in his right common iliac vein on the axial plane of abdominal C (To determine whether consistent with fat, Hounsfield unit measured at yellow Asterisk showed -86 Hounsfield units)

Global ischemic stroke

- Brain regions situated at **border zones** between major cerebral arteries: Anterior, middle, and posterior cerebral arteries particularly
- During **global hypoperfusion** → prone to **infarction**
- Cortical Watershed** (“watershed” is a drainage basin) **Zones:** Outer cortical areas where **ACA, MCA, and PCA territories meet**
- Subcortical Watershed Zones:** Deep brain regions, typically between **deep** vascular territories of MCA and ACA

FLAIR (Fluid-Attenuated Inversion Recovery) MRI suppresses bright signal of water in the CSF to make lesions with extra water (like inflammation or damaged tissue) stand out against the dark fluid

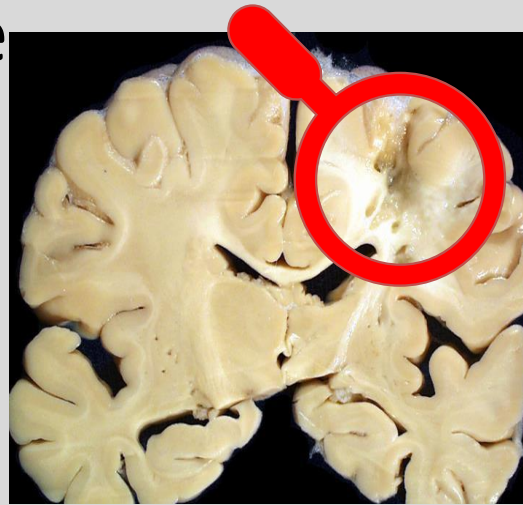


- <https://journals.plos.org/plosone/article/figures?id=10.1371/journal.pone.0108712>
- Watershed infarcts (WSI) territories indicated/ blue & red with FLAIR MRI: (a) Frontal region between anterior and middle cerebral arterial territories (b) Occipital WSI– between middle and posterior arteries (c) Internal WSI: Between superficial and deep perforators territories of middle cerebral artery (d) Paramedian WSI: Between anterior and middle cerebral arteries

Global ischemic stroke

- **Neurons** – are **VERY sensitive to hypoxia**
- **Most vulnerable: Hippocampus** (CA1 region), **neocortex** (layer 3, 5, and 6 pyramidal neurons, *insula*, precentral gyrus, and frontal lobes) cerebellum (**Purkinje cells**), **watershed areas (including reticular thalamus, caudate, putamen – deep locations** between artery territories → last to receive blood & high metabolic activity demands oxygen and glucose)
- Mnemonic: **Vulnerable hippos need pure water**
- If mild, only transient post-ischemic confusion with complete recovery
- If widespread neuron death & pt. survives → often persistent vegetative state

<https://lens.usercontent.google.com/image?vsrid=CN2EhdH1hsKFtAEOAhgBliRhZWY4MDRkMS1mYTMylTQwODgtOWY0NS05OTY1MzI4Yzc2OWUyBilCbm4oBTjd7NG-YzOSAw&gsessionid=mxw9C6ylbAnLHS-kXeC98-0jzaka8Y-ZzSAshb9W5PCQjTeenWT-Og>



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- <https://renaissance.stonybrookmedicine.edu/pathology/neuropathology/chapter2>
- Liquefactive necrosis at the paramedian watershed zone of the **neocortex**



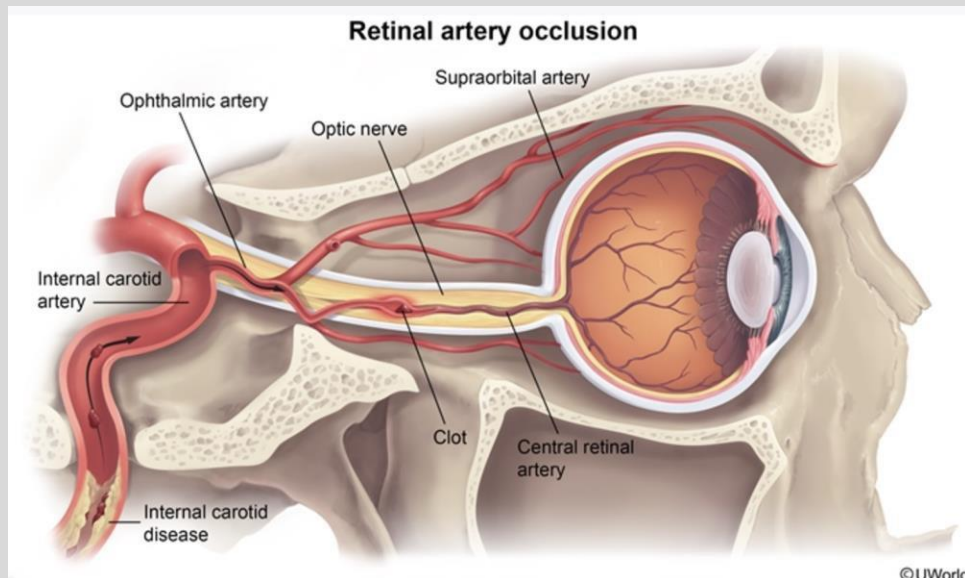
- “Insular cortex ribbon” ribbon loss, “insular ribbon sign” on left: Visible gray-white matter boundary of cortex loss is a key **early CT indicator of acute ischemic stroke** in **MCA** territory
- Vulnerable to hypoperfusion → rapid edema blurs normal gray and white matter boundary
- Right side insula cortex is highlighted and with arrows

Transient ischemic attack aka “mini stroke”

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Brief, reversible** episodes of **focal**, non- convulsive **ischemic neurologic disturbance**: Usually 2 – 15 minutes, may be longer (If symptoms disappear within 24 hours, event is termed “transient ischemic event/ attack” - symptoms same as full stroke, no permanent brain damage, to prevent a major stroke get immediate medical attention)
- Can experience many, and **between** attacks **may show normal neuro exam**
- Associated with **cerebral atherosclerosis**
 - **Amaurosis fugax**
 - Often a **warning sign** for a stroke or permanent blindness



©UWorld

Clinical management of ischemic stroke

NEW YORK INSTITUTE
OF TECHNOLOGY



Time Is Tissue Loss!

Recall BEFAST acronym of suspected stroke

B - Balance: Sudden dizziness, loss of balance, or trouble walking.

E - Eyes: Sudden trouble seeing in one or both eyes, or double vision.

F - Face Drooping

A - Arm Weakness

S - Speech Difficulty

T - Time to Call 911 (note the time symptoms started)

- https://www.itnonline.com/sites/default/files/styles/content_large/public/GettyImages-135138064.jpg?itok=0IS33-iQ
- Infarction's **deficits** determined by damages **anatomic distribution**, rather than underlying cause



Rx = medicine <4.5 hours of symptoms onset to dissolve clots, or surgical removal

- https://www.heart.org/en/-/media/Data-Import/images/7/9/8/0501-Feature-Stroke_WP.jpg?h=300&w=671&sc_lang=en
- Stroke/ Cerebrovascular Accident (CVA) results in **acute focal neurological symptoms**
- Symptoms* of a stroke can include **sudden** headache, blurry vision, dizziness, slurred speech, confusion, **contralateral** arm and face weakness (*facial drooping*) & numbness
- Thrombolytic therapy is **contraindicated** in a brain **hemorrhage** of any etiology: Rx= Control BP, reverse any anticoagulants, reduce brain swelling, repair underlying cause and relieve pressure

Ischemic stroke hemorrhagic transformation

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Sustained ischemia → **vessels** near infarct core **damaged** → **reperfusion** →

hemorrhagic conversion of infarct

- Or maybe, **ischemia-reperfusion injury** after spontaneous or therapeutic dissolution/ fragmentation of material
- Larger ischemic territories are more prone
- Types:

- **Hemorrhagic infarction: Petechial** bleeding into infarct **without mass effect**

- **Parenchymal hemorrhage:** More severe bleeding **with mass effect**, potentially life threatening



- HI-2 (Hemorrhagic Infarction type 2): More confluent petechial hemorrhages → no significant mass effect
- PH-2 (Parenchymal hematoma type 2): Hematoma occupies >30% of infarcted region → significant mass effect

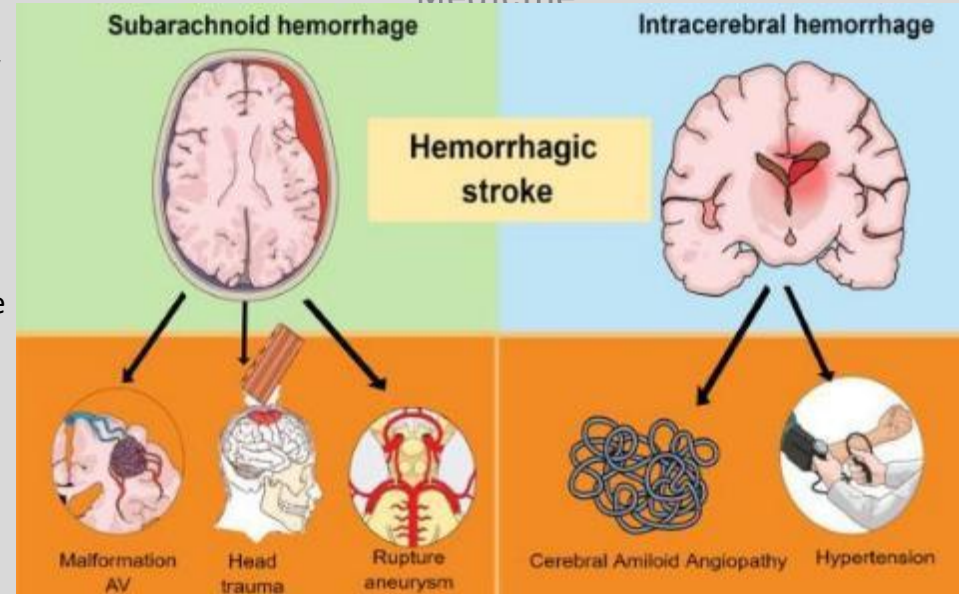


Hemorrhagic stroke

- Hemorrhagic stroke contributes to 10%-20% of strokes annually
- Hemorrhagic stroke is due to **bleeding** into the brain by the **rupture of a blood vessel** and is subdivided into **intracerebral hemorrhage (ICH)** & **subarachnoid hemorrhage (SAH)**
- **ICH** is bleeding directly into the brain tissue (parenchyma) and is commonly caused by **hypertension** or cerebral amyloid angiopathy (**CAA**)
- **SAH** is bleeding in the space around the brain – commonly caused by a **ruptured aneurysm**, **head trauma**, or an arteriovenous malformation (**AVM**)
- **More common in men** and increases with **age**
- In people on anticoagulant treatment, hemorrhagic infarcts may be extensive
- Venous infarcts are often hemorrhagic - may occur after thrombotic occlusion of superior sagittal sinus, other sinuses, or deep cerebral veins
- Epidural hemorrhage (EDH) & subdural hemorrhages (SDH) are types of intracranial bleeding & can cause stroke-like symptoms and rarely true ischemic stroke due to compression, but it's often misdiagnosed as stroke because it mimics symptoms

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

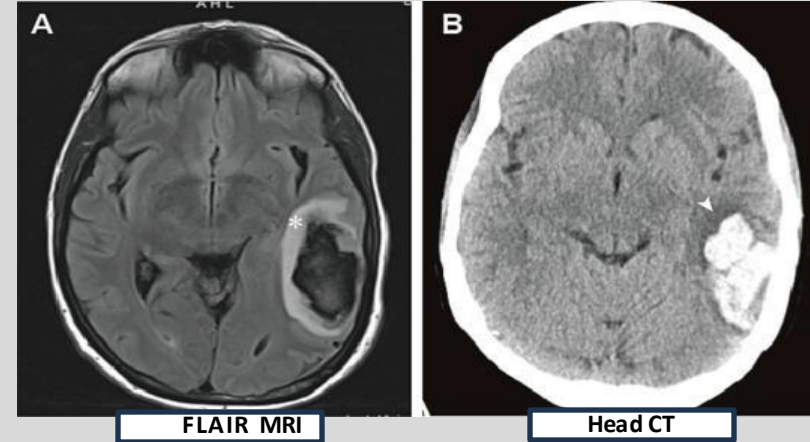


<https://www.sciencedirect.com/science/article/pii/S0146280622002250>

Hemorrhagic stroke pathophysiology

- Spontaneous (nontraumatic) intraparenchymal hemorrhages most common incidence ~60 years of age
- Hemorrhagic infarctions parallel ischemic with blood extravasation and resorption
- **Vessel rupture**
 - Hypertension, CAA, trauma, aneurysm or vascular malformations
 - **Blood extravasates** into brain parenchyma or subarachnoid space
- **Mass effect & edema** (see, asterix & arrowhead)
 - Hematoma **compresses surrounding brain** (hyperdense area on **CT**, usually in brain tissue)
 - Results in **elevated intracranial pressure** (ICP)
- Acute hemorrhages have central core of clotted blood that compresses adjacent parenchyma → secondary infarction of the affected brain tissue, with anoxic neuronal and glial changes as well as edema (eventually resolves) → hemosiderin and lipid-laden macrophages appear & proliferation of reactive astrocytes at periphery
- **Secondary ischemia & stroke**
 - **Adjacent vessels constrict** → worsens tissue hypoxia with **secondary ischemic strokes**
- **Toxic effects of blood products**
 - Hemoglobin breakdown releases free iron and reactive oxygen species (**ROS**) → cellular injury, BBB disruption, and neuroinflammation

NEW YORK INSTITUTE
OF TECHNOLOGY



- Neurobiol Dis. 2023 Jan;176:105948
- Brain **edema** formation and therapy after ICH
- (A) ICH patient's 2 days MRI FLAIR after hemorrhage showing significant **hyperintense perihematomal edema** (white asterisk) & lateral ventricle compression (B) ICH patient's 1 hour CT after hemorrhage showing **low-density brain edema** region (white arrow) around the hematoma

Intracerebral hemorrhage: Chronic hypertension

- Hypertensive change causes **non-lobar (rather than cerebral lobes)** intracranial hemorrhage (**ICH**)
- Hypertension vessel wall abnormalities include accelerated atherosclerosis in larger arteries, hyaline arteriosclerosis in smaller arteries – in severe cases, frank necrosis of arterioles
- If small vessels with arteriosclerosis do not rupture but are occluded → lacunar infarction
- Long standing hypertension → **Charcot-Bouchard microaneurysms** (due to degeneration of tunica media, elastic lamina & smooth muscles, lipohyalinosis, subendothelial fibrinoid necrosis, microaneurysms) – **commonly** find bulge in a **small penetrating arteries** supplying **basal ganglia** (originate from basilar arteries or anterior, middle, or posterior cerebral arteries (eg, **lenticulostriate arteries**)
- Acute hypertension can also cause ICH, ex. Eclampsia - known as postpartum ICH



Intracerebral hemorrhage: Cerebral amyloid angiopathy (CAA)

- Caused by accumulation of **cerebral amyloid- β (A β)** in **tunica media** and **adventitia** of **leptomeningeal & cortical vessels**
- Amyloid deposits lead to inflammation, causing lesions, hemorrhages - requires immunosuppression for inflammation
- Causes **vascular fragility** tends to manifest in **normotensive elderly** patients as **lobar** intracerebral hemorrhage
- CAA $\uparrow$ **age**, ~50% >70 years have CAA - **recurrent** hemorrhages can occur due to CAA
- Associated with variations in the gene encoding **apolipoprotein E** (either an $\epsilon 2$ or $\epsilon 4$ allele, $\uparrow$ risk of bleeding)

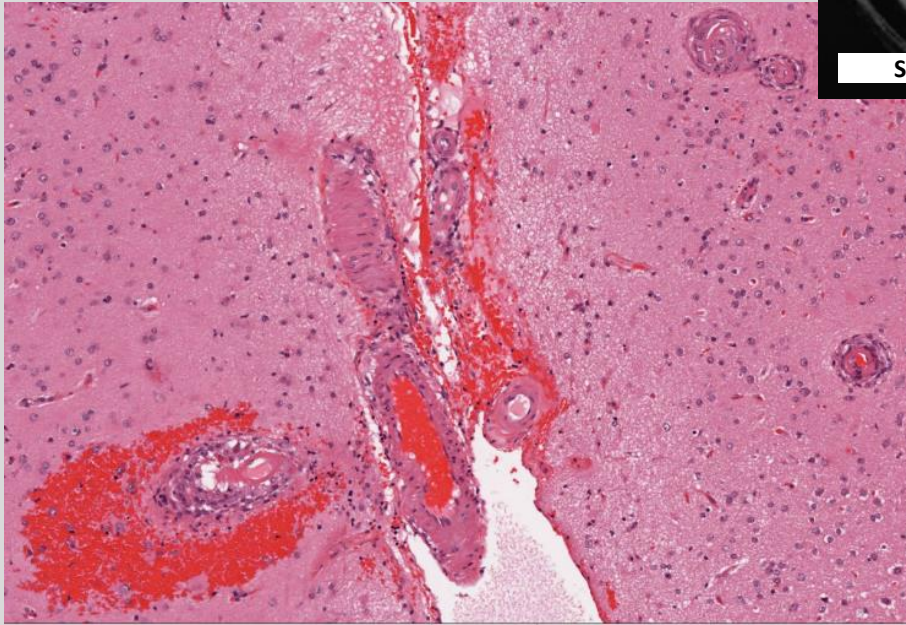
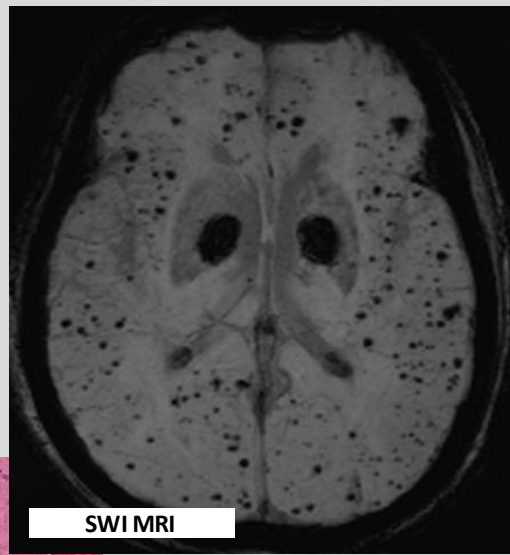


CT Head

https://researchmgt.monash.edu/ws/portalfiles/portal/321744859/292856193_oa.pdf

CAA & ICH

- Pathology similar to Alzheimer's disease (AD), however in AD amyloid deposits in brain parenchyma
- Patients with CAA have evidence of numerous **microhemorrhages** within brain ("microbleeds")
→ may cause **vascular dementia**



- https://lmping.med.utoronto.ca/LMP/36980.htm?utm_campaign=Full&utm_medium=referral&utm_source=dml
- **Beta amyloid angiitis**
- Immunohistochemical staining for A β (beta amyloid) will show **deposited in vessel walls**

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- <https://radiopaedia.org/cases/pres-and-cerebral-amyloid-angiopathy>
- SWI MRI **technique** highly sensitive to blood/ blood products, iron, and calcium to visualize **microbleeds**, veins, and calcifications - for diagnosing **neurodegenerative** diseases, **strokes** & traumatic brain injuries
- In **SWI MRI**, foci of SWI susceptibility artifact in both cerebral and cerebellar **hemispheres** with **sparing** of the **basal ganglia, thalami & brainstem** - more consistent with **cerebral amyloid angiopathy**
- **Susceptibility Weighted Imaging SWI MRI**: highlights blood products (hemosiderin) as **blooming artifacts** (in medical imaging some lesions appear larger than actual) showing **dot hemorrhages** (pure water or fluids like CSF generally appear as a **dark** (low signal intensity) signal)

Subarachnoid hemorrhage (SAH)

- In the **subarachnoid space** (CSF circulates here) - Imaging: Blood in the subarachnoid space on CT
- Most commonly from **trauma**, then **ruptured aneurysms** (e.g., **berry aneurysms**)
- Presents as a **sudden** onset of a **severe “thunderclap” headache** (patients may say “this is the **worst headache** of my life” reaching maximum intensity in minutes), **neck stiffness**, **photophobia** and can rapidly **lose consciousness**



Normal head CT



**NEW YORK INSTITUTE
OF TECHNOLOGY**

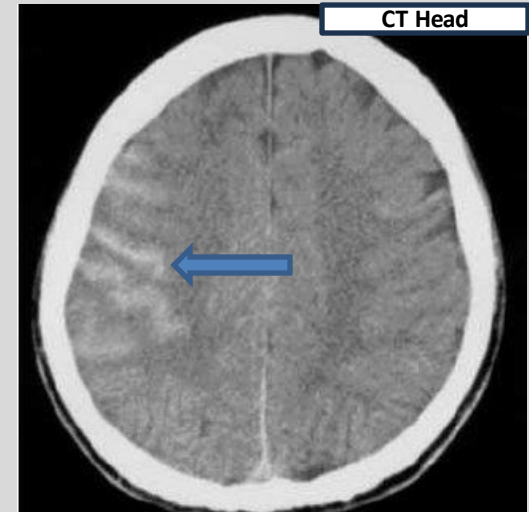
College of Osteopathic
Medicine

Head CT showing hemorrhages
arrow: Ruptured aneurysmal SAH
in a starburst pattern (blood in
basal cisterns), but also right
sylvian fissure (Classic pattern of
SAH with high density blood
surrounding the **circle of Willis**,
where majority of **berry**
aneurysms are located)

<https://commons.wikimedia.org/wiki/File:SubarachnoidP.png#/media/File:SubarachnoidP.png>

SAH: Trauma commonly produces sulcal SAH (see head CT arrow)

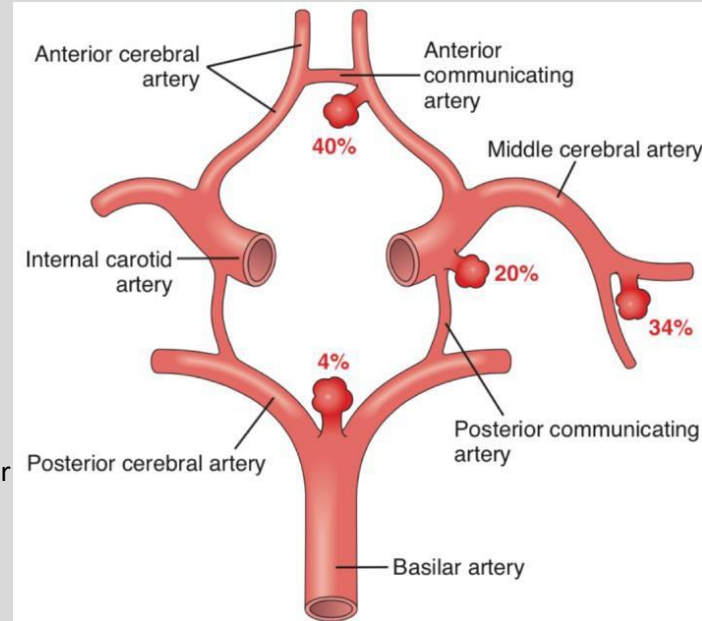
- **Most common overall** cause of SAH
- Classically presents **with blood in the sulcus** of the brain due to **tear of small arteries** travelling in **subarachnoid space**
- **Often occurs near bony regions**, like over the skull base as brain hits against bone from trauma



- https://www.researchgate.net/figure/Traumatic-subarachnoid-hemorrhage-Axial-non-enhanced-CT-shows-high-density-hematoma-in_fig4_7426978
- Non-contrast enhanced head CT with high density blood in cortical sulci (arrow)

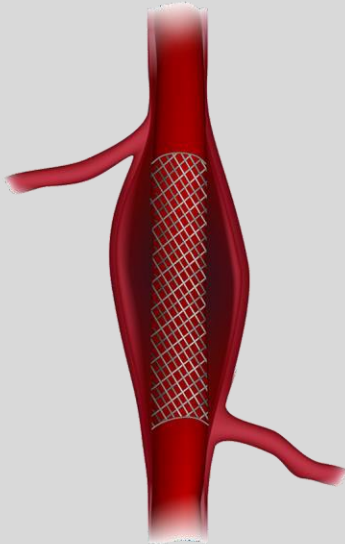
SAH: Spontaneous - aneurysm

- Most common *spontaneous* SAH is **aneurysmal**
- Most commonly **saccular**, aka "**berry**" aneurysms of the **Circle of Willis** (in subarachnoid space)
- **Berry aneurysms** - small, thin-walled protrusions of usually arteries of circle of Willis or major branches
- **Majority** are on **anterior circle of Willis** (**anterior communicating** artery most common – see diagram)
- In ~ 2% of population; multiple in 20-30% of cases in autopsy series
- Risk factors
 - Collagen/Elastin disease (ex. **Marfan's** & **Ehlers Danlos** syndromes)
 - **Female**
 - **Family history**
 - **Smoking**
 - Autosomal dominant polycystic kidney dz
- Aneurysms are few millimeters to 2 or 3 cm in diameter, bright red, shiny surface and thin, translucent wall - risk of bleed ↑ with size: >10 mm diameter have ~50% risk of bleeding per year
- May rupture any time, but ~1/3rd of cases acute ↑ICP, like straining at stool or sexual orgasm
- 25-50% of patients die with first rupture, but survivors often improve and recover consciousness in minutes - repeat bleeding is common with worse prognosis each time
- Other aneurysm types include atherosclerotic (fusiform; mostly of the basilar artery), mycotic, traumatic, and dissecting but more often cause cerebral infarction than SAH

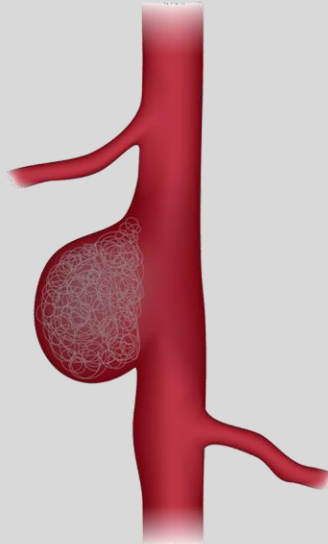


Clinical management of cerebral aneurysms include the following:

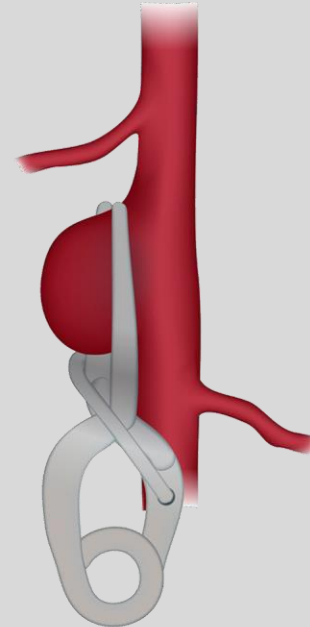
Stenting



Coiling

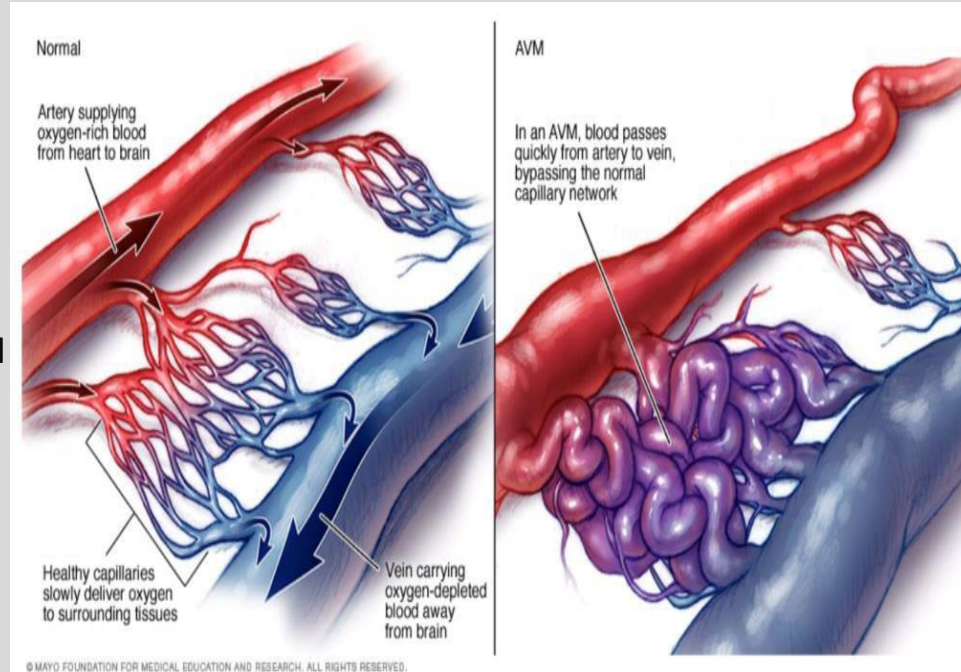


Clipping



SAH: Spontaneous- (arteriovenous malformations)AVM

- Important cause of spontaneous **SAH**, but can **also result in intracerebral hemorrhage**
- Region of **dysplastic dilated blood vessels** (neither arteries nor veins) **arterial** blood flows directly into draining **veins without** normal interposed **capillary beds**
- **Usually congenital** and **enlarge** with age, and often progress from **low-flow juvenile** lesions at birth to **medium-to-high-flow high-pressure** lesions in **adulthood**
- Although lacking features of typical neoplasms, AVMs frequently have activating mutations in KRAS oncogene in endothelial cells
- Grossly are **“tangle of vessels”** in subarachnoid space, brain, or both



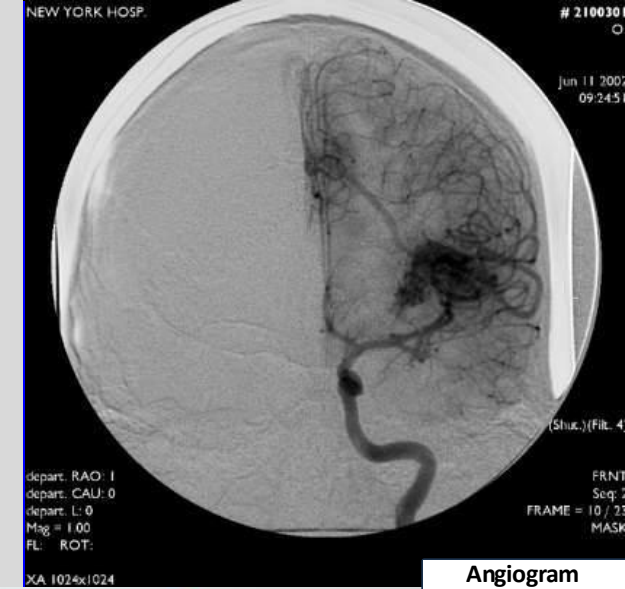
© MAYO FOUNDATION FOR MEDICAL EDUCATION AND RESEARCH. ALL RIGHTS RESERVED.

AVM blood flow

In an arteriovenous malformation (AVM), blood passes quickly from the artery to vein, disrupting the normal blood flow and depriving the surrounding tissues of oxygen.

SAH: Spontaneous- AVM

- **Males : female = 2:1**
- Often presents between 10-30 yrs of age as a **seizure** disorder, **intracerebral hemorrhage**, or **SAH**
- Most common site is **middle cerebral artery** territory, especially posterior branches
- **Management** of **unruptured** AVM involves **surgical** resection based on Spetzler- Martin Grade (size, location, venous drainage)



Angiogram

Spetzler-Martin AVM Grading Scale	Points
Size	
0-3 cm	1
3.1-6.0 cm	2
> 6 cm	3
Location	
Noneloquent	0
Eloquent *	1
Deep venous drainage	
Not Present	0
Present	1
AVM Total Score	1-5

The Spetzler-Martin AVM Grading Scale is based on size, location, and venous drainage of intracerebral AV malformation. The score is calculated by adding the points for each category. The range is 1 to 5. The lower the score, the better the outcome.

***Eloquent locations-**
areas of sensorimotor, language, visual, thalamus, hypothalamus, internal capsule, brain stem, cerebellar peduncles, and deep cerebellar nuclei

<https://radiopaedia.org/articles/spetzler-martin-arteriovenous-malformation-grading-system-2?lang=us>

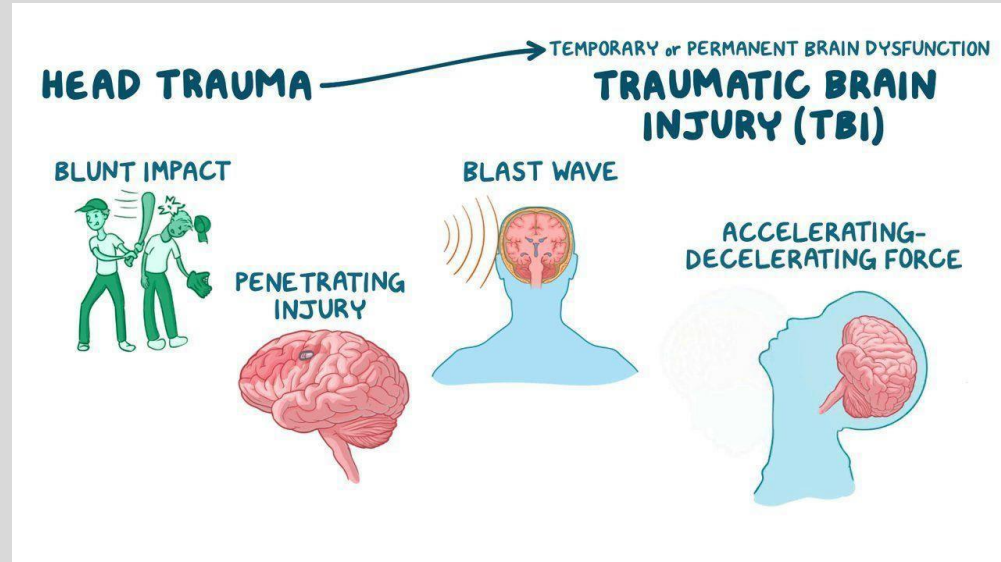
DO NOT MEMORIZE!

CNS trauma

- **Physical forces** causing head injury may cause **skull fractures**, **parenchymal injury** & **vascular injury**, or all three
- Can include the following types of forces: **Penetrating**, **blunt**, **blast**, **accelerating-decelerating** - either **open** or **closed injury**
- Multiple etiologies – automobiles, abuse, falls, bullets, etc.
- Head injury/trauma is leading cause of **death** - **ages 15 – 40 yrs**
- Major cause of **permanent disability**
- **Most common** cause of **pediatric traumatic death**

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



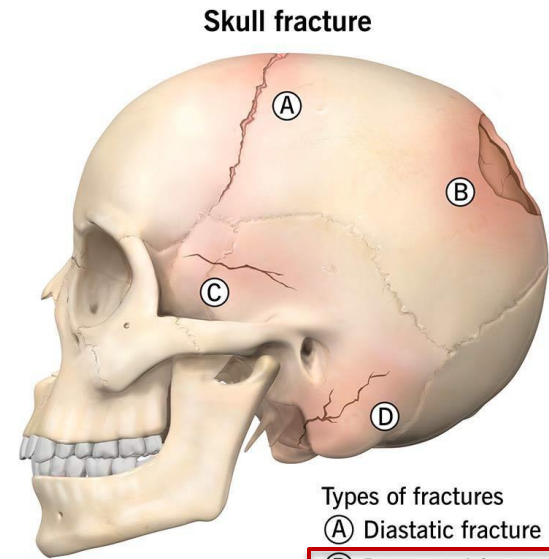
<https://www.rxteach.com/emergency-tbi-management/>

Skull fractures: Depressed type – types & mgmnt.

- Depressed skull fracture cause **compression of bone into brain**: Parenchyma or cortical vessels
- Surgery** is considered for depressed skull fractures with **deficits** referable to **underlying brain**
- Distance greater than thickness of bone = displaced skull fracture



- <https://learningradiology.com/notes/generalnotes/depressed%20skull%20fx.htm>
- CT of the brain in bone window with **depressed skull fracture** of right **parietal bone** (red arrows) with one part of fracture depressed beneath inner table of the other fragment
- White arrow shows soft tissue **swelling**
- Bone window: Contrast is set to highlight dense tissues like bone
- Diastatic = suture separation



Types of fractures

(A) Diastatic fracture

(B) Depressed fracture

(C) Linear fracture

(D) Basilar fracture



<https://my.clevelandclinic.org/health/diseases/skull-fracture>

Skull fracture: Basilar

- Typically see **orbital roof**, **sphenoid bone**, or **petro-mastoid portion of temporal bone fractures** (usually occiput or sides of head impact)
- Typical signs: **Battle's sign** - ecchymosis behind ear (mastoid process), **raccoon's eye** signs (blood from skull fracture seeps into soft tissue around eyes)
- **CSF leakage** from the nose or ear (rhinorrhea/otorrhea) and infection (**meningitis**) may follow



<https://neupsykey.com/neurologic-history-examination/>



- <https://www.jetir.org/papers/JETIR1901762.pdf>
- Head axial non-contrast CT Head shows linear fracture involving foramen magnum

NEW YORK INSTITUTE OF TECHNOLOGY

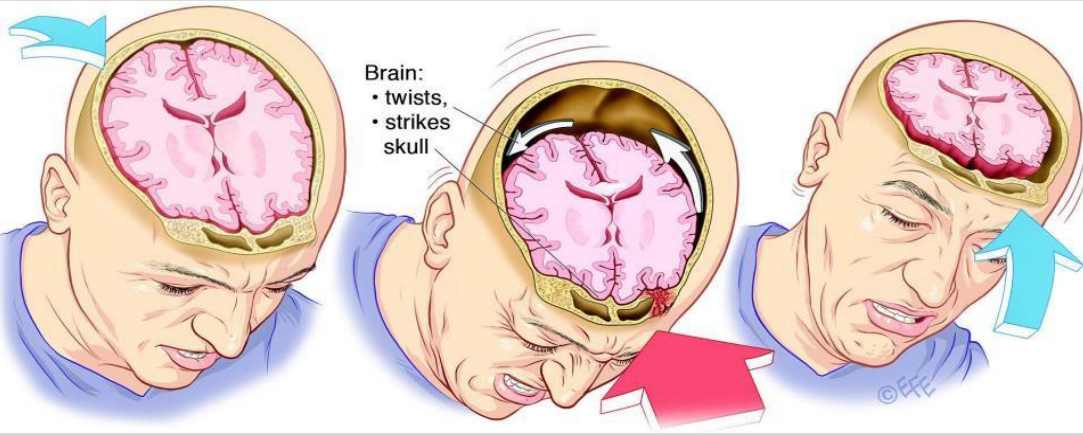
College of Osteopathic
Medicine



<https://neupsykey.com/neurologic-history-examination/>

Traumatic brain injury (TBI)

- Brain injury from trauma results from **two distinct processes**: **Primary brain injury** at time of trauma (**cortical contusions**, lacerations, bone fragmentation, **diffuse axonal injury**, and **brainstem contusion**); **Secondary injury** after **initial injury** from **intracranial hematomas**, **edema**, hypoxemia, ischemia (primarily due to **elevated intracranial pressure (ICP)** and/or shock), vasospasm
- Concussion** is a functional manifestation of **mild** traumatic brain injury



NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

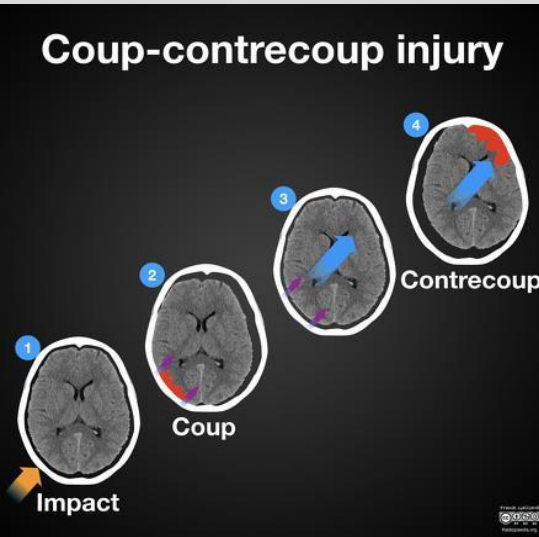
- <https://standardofcare.com/concussion/#:~:text=A%20sport%20related%20concussion%20is%20a%20traumatic,psychological%20implications%20in%20both%20short%20and%20long%20term>
- Concussion: Clinical syndrome biomechanical induced alteration of brain function (typically affects **memory & orientation**, *may* involve loss of **consciousness**)
- Typically with **immediate & transient** loss of consciousness with a brief period of amnesia following force to the body (including head, rapid brain movement causes axonal stretching and microstructural damage) that effects the brain - neurologic **recovery usually complete**
- Diagnosis based on history & physical: Acute Sx - Headache, dizziness, confusion, nausea, blurred vision; delayed Sx - mood changes, memory issues, sleep disturbances
- Functional manifestation of mild traumatic brain injury, but **no** trauma related intracranial **abnormality** is seen on **CT or MRI**
- Neurometabolic cascade causes brain to be more **sensitive** to stress and further injury **until fully recovered**

Coup/contre-coup injury

- Blow to head may result in **contusion at point of contact (coup injury)** or on **brain surface diametrically opposite it (contrecoup injury)**
- **Coup injury** = injury at point of contact/impact (head generally immobile at time of trauma)
- **Contre-coup injury** = damage to brain surface opposite of impact

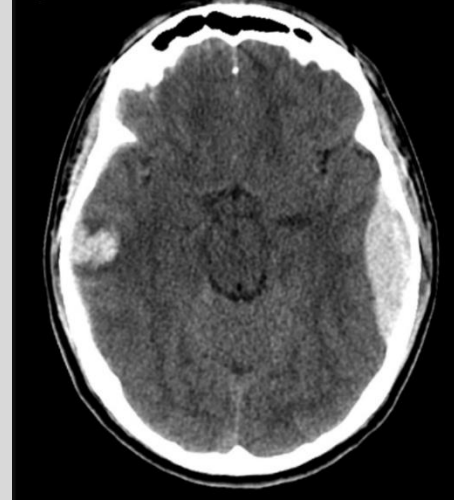
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

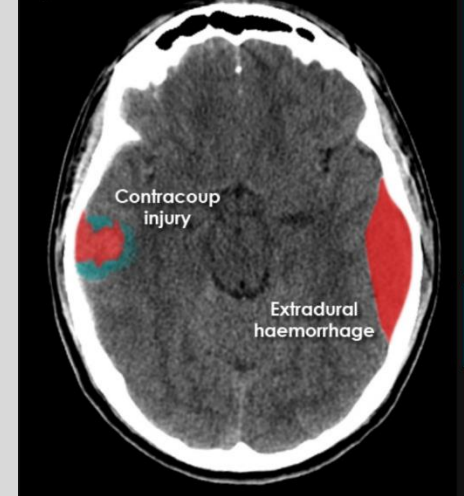


<https://radiopaedia.org/articles/coup-contrecoup-injury-brain>

Extradural haematoma (EDH) with contrecoup injury
Right Left



Extradural haematoma (EDH) with contrecoup injury
Right Left



- https://www.radiologymasterclass.co.uk/gallery/ct_brain/ct_brain_images/extradural_contrecoup_ct_brain_top_1st_img
- Brain CT shows large **left acute epidural** hematoma & contrecoup injury on the right keeping with a coup-contrecoup injury pattern

Direct parenchymal injury

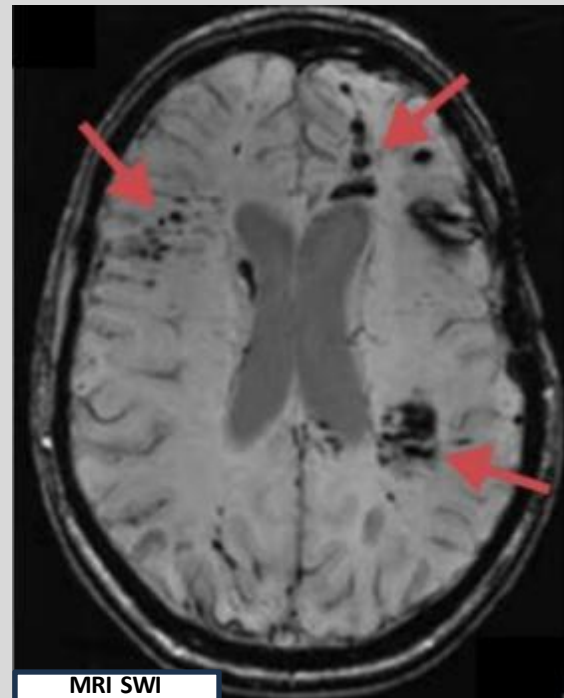
- **Contusion: Bruise** caused by blunt trauma
 - Wedge-shaped, broad base along surface at impact point
 - **Blow to surface** of brain, **transmitted** through skull, leads to **rapid tissue displacement**, disruption of vessels, hemorrhage, tissue injury & edema
 - Manifestations of parenchymal injury range from mild (concussion) to severe (contusions and lacerations)
- Postconcussive neuropsychiatric syndromes, typically associated with repetitive injuries - chronic traumatic encephalopathy (CTE, previously “dementia pugilistica”), a dementing illness after repeated head trauma
- CTE brains are atrophic, enlarged ventricles, accumulation of tau- neurofibrillary tangles in gyri and perivascular in frontal and temporal lobe cortices
 - Important sequelae of brain trauma include posttraumatic epilepsy (focus is healed injury site), risk of infection, posttraumatic hydrocephalus, and psychiatric disorders



- https://commons.wikimedia.org/wiki/File:Brain_trauma_CT.jpg#:~:text=Multiple%20frontal%2C%20parietal%2C%20and%20temporal%20lobe%20contusions,was%20noted%20without%20shift%20of%20midline%20structures.%22
- Bilateral **frontal lobe damage**: Areas of high density (white) both sides of frontal lobes indicate **contusions (bruising)**

Diffuse axonal injury (DAI)

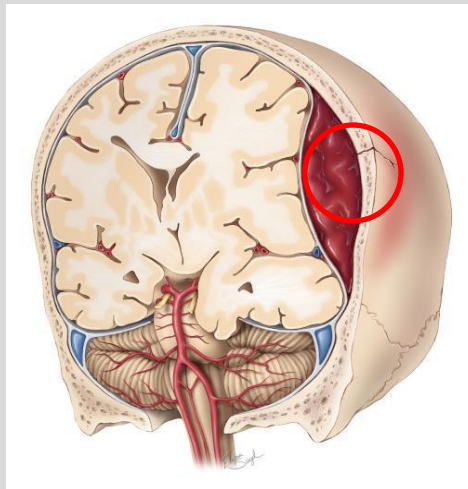
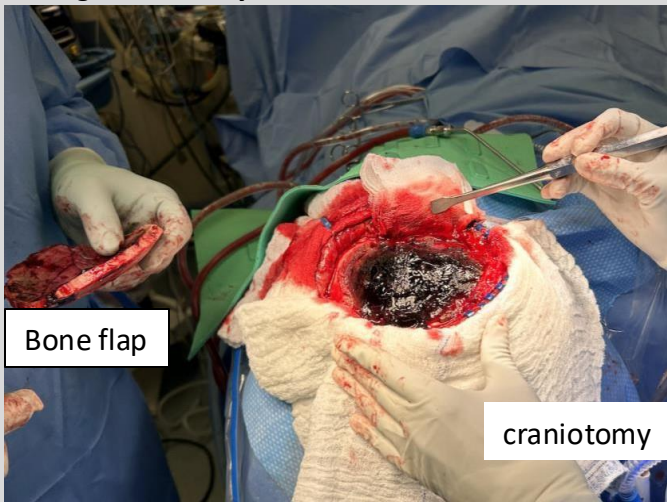
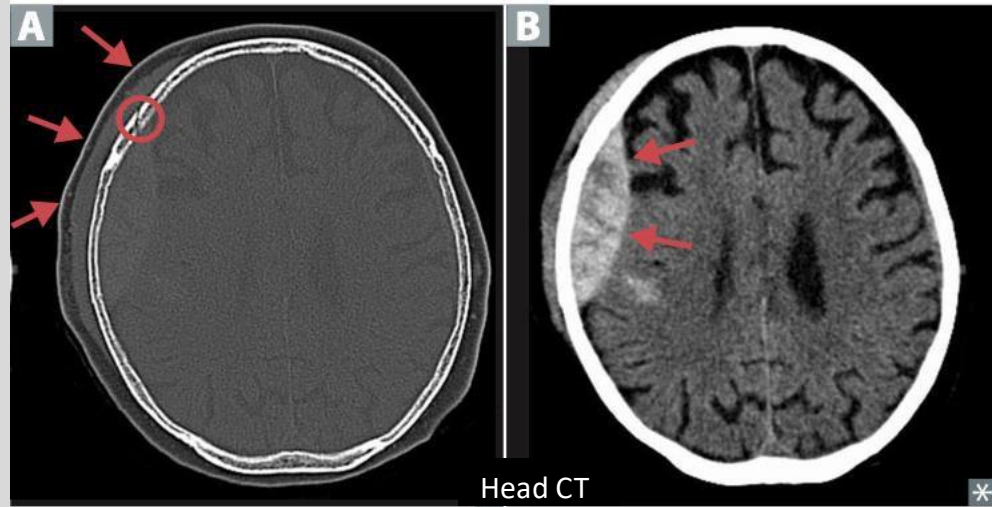
- Traumatic brain injuries **damage deep white matter regions**, cerebral peduncles, superior colliculi, and deep reticular formation in brainstem
- Widespread brain injury from **acute stretching & shearing forces** during trauma, damaging nerve fibers (**axons**)
- Microscopic findings include axonal **swelling**, and **focal hemorrhagic** lesions subsequently there's degeneration of involved fiber tracts
- Imaging studies show **petechial hemorrhages** in white matter tracts
- **High speed brain injury** may occur from marked changes in **angular acceleration** (e.g., **blast injuries, MVC**), even in **absence of physical impacts of skull**
- Associated with **immediate, deep coma**
- Cerebral **edema** and **↑ICP**
- Mortality 30-40%



- [magnetom_flash_ismrm_issue_62-02161984_1800000002161984.pdf](#)
- For detecting tiny hemorrhages (**microbleeds**) and axonal damage invisible to CT, especially in white matter, brainstem, and corpus callosum - **SWI** is highly sensitive for detecting these hemorrhagic lesions, which are key indicators of DAI
- Susceptibility-weighted magnetic resonance imaging (**MRI**) scan of the brain, highlighting **multiple cerebral microbleeds**
- Dark spots (red arrows) represent hemosiderin deposits from old, small hemorrhages

Epidural hematoma (EDH)

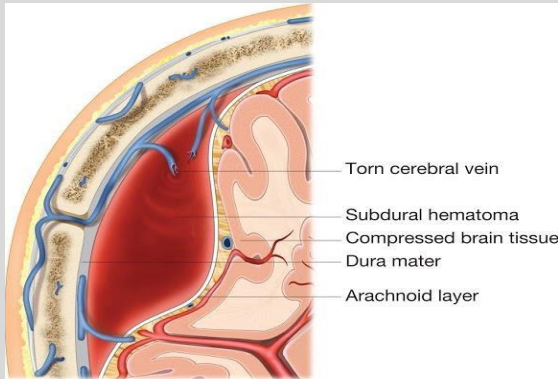
- Located **between skull** and periosteal layer of **dura**
- Often due to **trauma**, particularly with skull fractures that damage the **middle meningeal artery – arterial blood**
- **Presents** usually with “**lucid interval**” followed by **rapid deterioration** - symptoms include headache, vomiting, and potential loss of consciousness
- **Prompt diagnosis** needed, **surgery for drainage** - fatal brain **herniation** may occur in a few hours - neurosurgical **emergency** that most often requires **evacuation** of hematoma using **craniotomy**



- <https://quizlet.com/64534757/first-aid-cns-part-three-flash-cards/>
- **Biconvex, lentiform-shaped extra-axial fluid collection characteristic** of an epidural hematoma
- <https://www.neurosurgicalatlas.com/volumes/emergency-neurosurgery-and-trauma/traumatic-hematoma/epidural-hematoma>
- Almost always associated with **skull fracture** (red circles)
- Arrows show soft tissue swelling & EDH

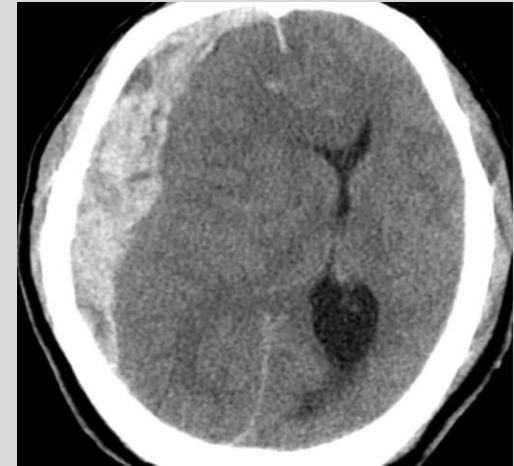
Subdural hematoma (SDH)

- Located **between dura and arachnoid mater**
- Usually due to **tearing of bridging veins (venous blood) often** from **head injuries** (often in **elderly** and **chronic alcoholics** (brain atrophy) associated with minor trauma or falls→ **chronic subdural**)
- Usually, venous bleeding is **self-limited** and the resulting hematoma is broken down and organized - slowly progressive neurologic deterioration is typical, but acute decompensation may also occur
- **Organized hematoma** is firmly attached to dura by fibrous tissue and can eventually retract as granulation tissue matures until thin layer of connective tissue remains - multiple recurrent episodes of bleeding may occur (chronic subdural hematoma), presumably from the thin-walled vessels of granulation tissue and re-injury
- Symptoms, may develop over hrs-wks, include headache, confusion, & focal neurological deficits -risk of repeat bleeding is greatest in first few months after initial hemorrhage
- Treatment of subdural hematomas = remove blood and organizing tissue
- At surgery, **acute** SDH hematoma looks like grape jelly, **chronic** SDH has motor oil appearance & SDH forms a capsule around hematoma (neovascularization)



**NEW YORK INSTITUTE
OF TECHNOLOGY**

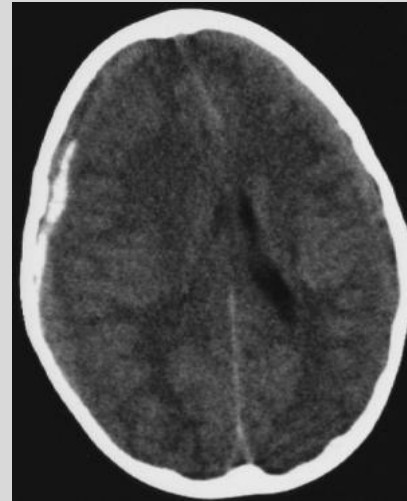
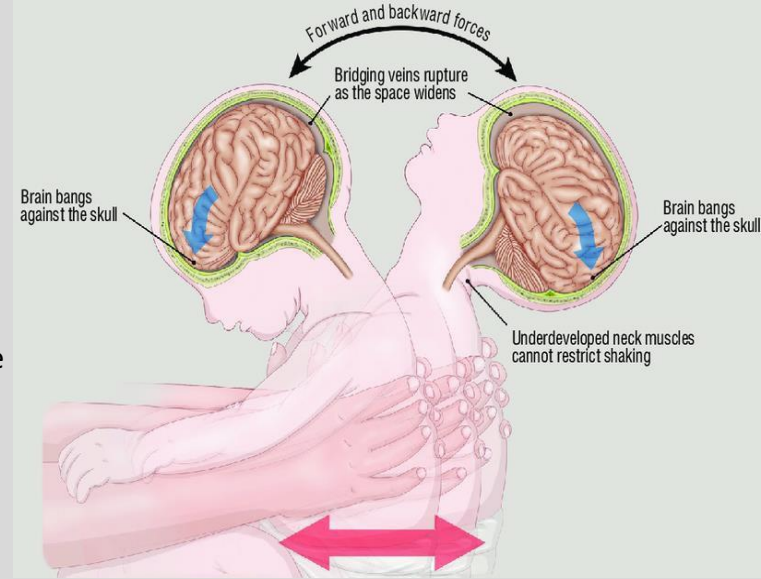
College of Osteopathic
Medicine



- <https://radiopaedia.org/cases/acute-subdural-haemorrhage-with-subfalcine-herniation>
- Imaging shows **crescent-shaped (concave)** bleed on **CT scan**
- Large right frontal **acute subdural** hematoma with significant mass effect, left midline shift and **subfalcine herniation**
- **Crescent-shaped bright (hyperdense) blood** on right side
- A **shift of midline** structures of the brain to the left
- **Compression** of underlying brain tissue

SDH - shaken baby syndrome

- <https://www.saskoer.ca/ece142hsn2ed/chapter/chapter-six-child-maltreatment/>
- A severe form of **child abuse** caused by violent shaking of an infant or young child - **neck muscles are not strong enough** to restrict head movement
- **Brain bangs against skull/ tearing of bridging veins** → SDH/ can also cause diffuse axonal injury
- Sx/Sx: Lethargy, seizures, apnea, **retinal hemorrhages** (>80% of cases)
- Associated findings a **history of many fractures** (e.g., ribs, long bones)



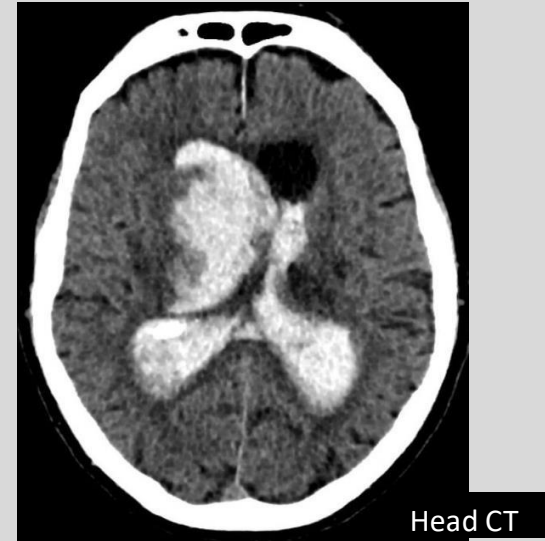
- <https://www.hawaii.edu/medicine/pediatrics/pemxray/v5c07.html>
- **Acute subdural** hematoma **right fronto-temporal** region
- **Compression** of right lateral ventricle with **shift** of midline structures from right to left
- Hematoma is **crescent shaped** as classically described with subdural hematomas

Intraventricular hemorrhage

- **Located** within the brain's **ventricular system**
- Can result from **extension of intracerebral hemorrhage** or **trauma**
- Commonly seen in **preterm infants (germinal matrix** - transient, highly vascular layer in the developing fetal brain, located near the ventricles, that produces neurons and glial cells - **hemorrhage)**
- Presentation is variable - may cause symptoms of **hydrocephalus** if blood obstructs CSF flow
- **Imaging** shows **blood in ventricular spaces** on CT

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



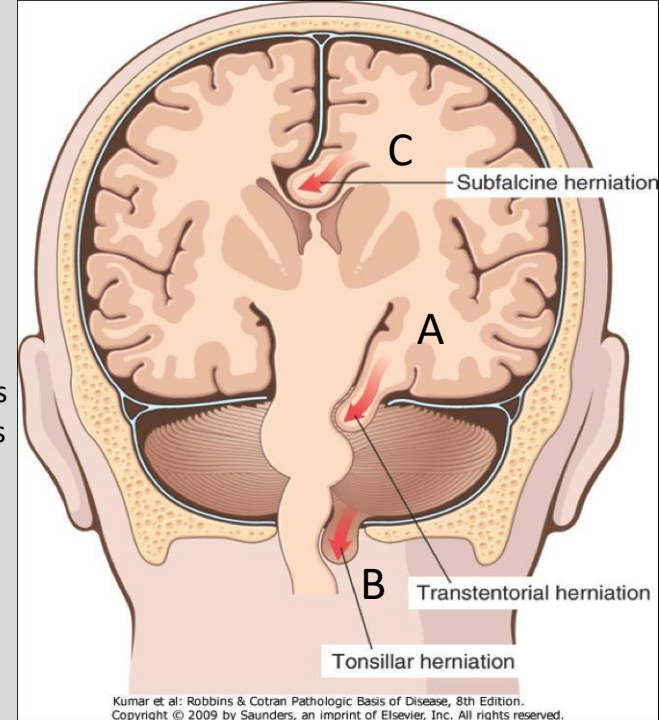
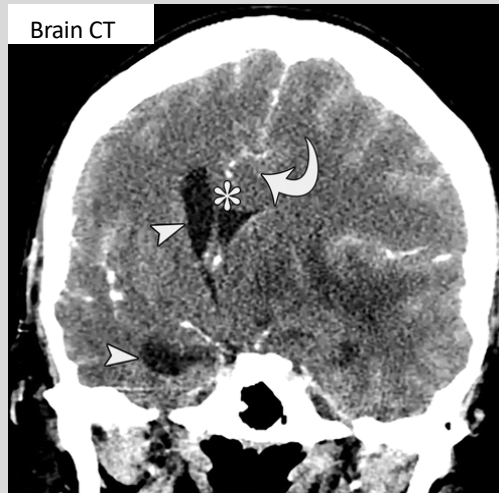
Head CT

- <https://pacs.de/term/intraventricular-hemorrhage#&gid=1&pid=13>
- Basal ganglia hemorrhage with intraventricular extension

Brain herniations

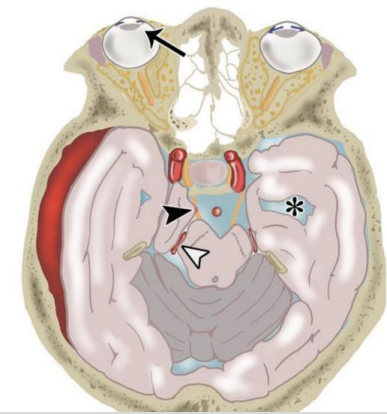
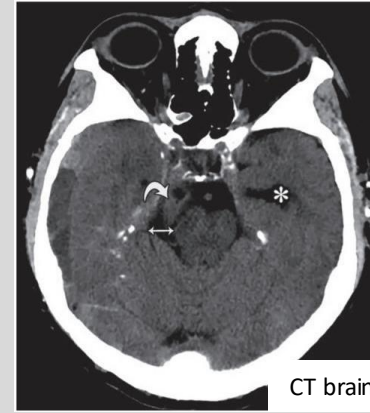
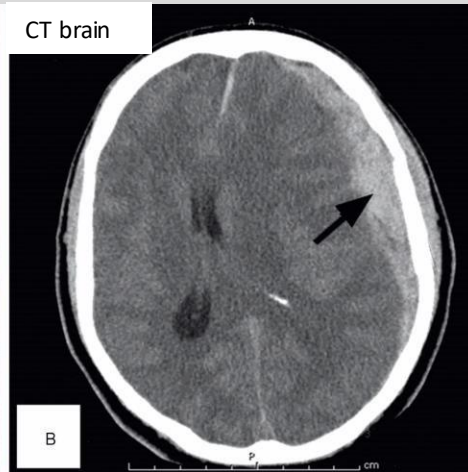
- A brain herniation occurs **when something inside the skull produces pressure** that shifts brain tissues
- Most often result of brain swelling or **bleeding** from a head injury, **stroke** or **tumor**
- Types: A. **Transtentorial (uncal)** herniation: Medial portion of temporal lobe – uncus herniates **down through tentorium cerebelli**; B. **Transforaminal (tonsillar)** herniation: Cerebellar tonsils herniate **down through foramen magnum**

- Riveros Gilardi B, Muñoz López JJ, Hernández Villegas AC, Garay Mora JA, Rico Rodríguez OC, Chávez Appendini R, De la Mora Malvárez M, Higuera Calleja JA. Types of Cerebral Herniation and Their Imaging Features. Radiographics. 2019 Oct;39(6):1598-1610. doi: 10.1148/rg.2019190018. PMID: 31589570C
- C. **Subfalcine/ cingulate** herniation: Cingulate gyrus herniates down **below falx cerebri (*)**, **midline shift** of ventricles (arrowheads)
- Most common type
- Can cause **headaches** due to **increase intracranial pressure**, also **contralateral leg weakness** (herniating cingulate gyrus can compress ipsilateral anterior cerebral artery against the falx → infarction (stroke) - ACA's supplies motor cortex for lower limb)



Transtentorial herniation

- Uncus compresses **ipsilateral oculomotor nerve**: **Ipsilateral fixed & dilated pupil, "blown pupil"** – see image A below
 - Need immediate neurosurgical intervention**: Craniectomy to evacuation of hematoma and relieve pressure



- <https://pubs.rsna.org/doi/abs/10.1148/rq.2019190018>
- Subdural hematoma, **right pupil was dilated and unresponsive to light**
- Axial CT image: **Uncus displaced down** across tentorial incisura (curved arrow)
- Ipsilateral perimesencephalic cistern** (double-headed arrow) is **widened**
- Contralateral cistern is compressed**
- Also contralateral dilatation of the temporal horn of the lateral ventricle (*)
- Compression of **posterior cerebral artery** (white arrowhead) & **oculomotor nerve** (black arrowhead, **ipsilateral blown pupil** (arrow) and contralateral temporal horn dilatation (*)

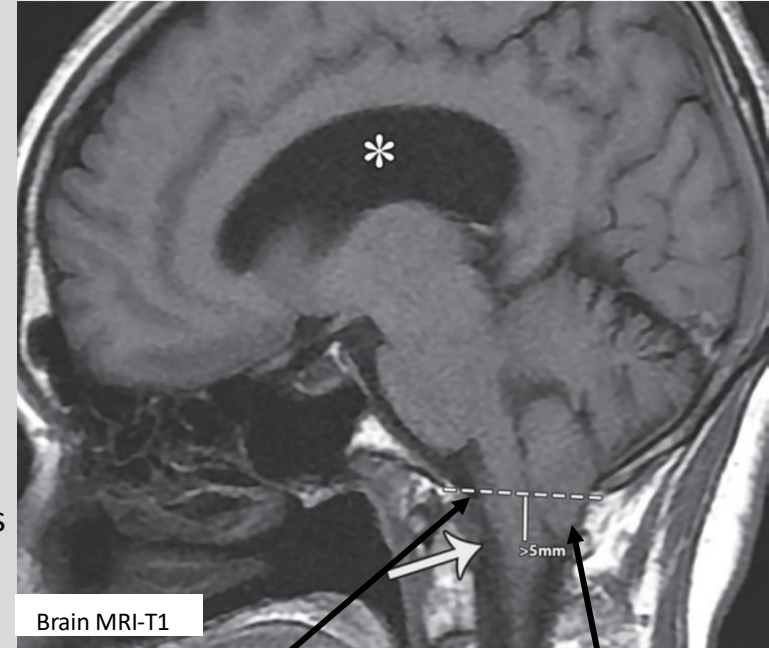
Demetriades D, Benjamin ER. Head Injury. In: Demetriades D, Chudnofsky CR, Benjamin ER, eds. *Color Atlas of Emergency Trauma*. Cambridge University Press; 2021: 1-23.



- Blown pupil**: **Large pupil** that **doesn't react to light at all** (fixed mydriasis) is a serious sign
- Normal in Bright Light: 2-4 mm
- Normal in Dim Light: 4-8 mm

Transforaminal/ tonsillar herniation

- **Cerebellar tonsils** protrude **through foramen magnum** due to **↑ICP**, often from conditions like TBI, hemorrhage, or tumor
- **Compression of brainstem & medulla oblongata**, disrupting vital functions such as respiration and cardiovascular regulation
- **Obtundation** → Neurosurgical **emergency** with often very poor prognosis

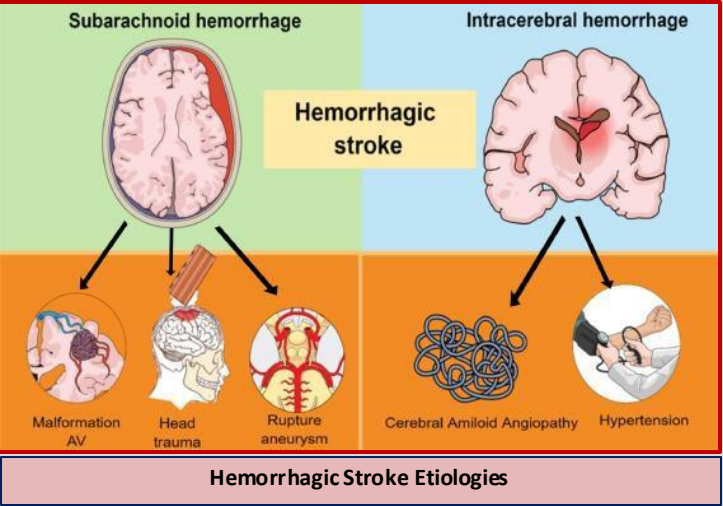
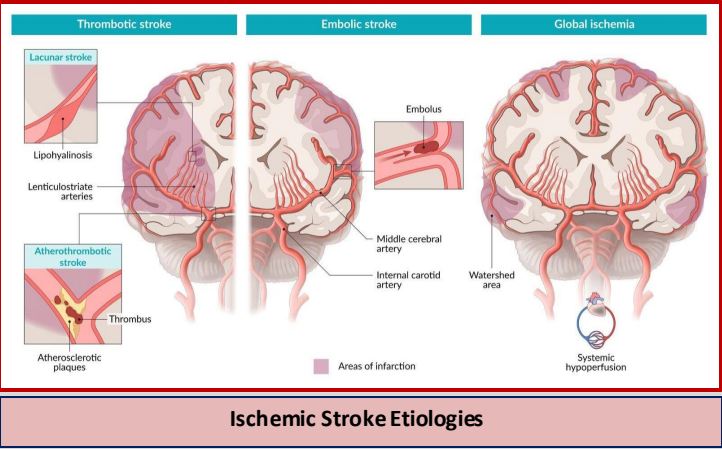


Foramen magnum

Tonsillar herniation

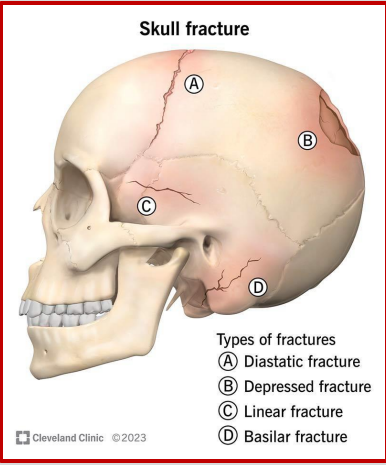
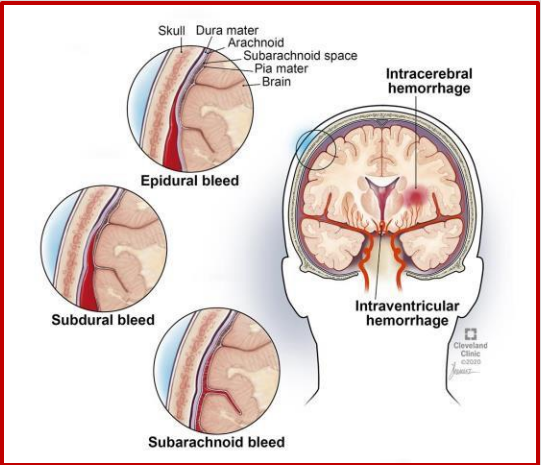
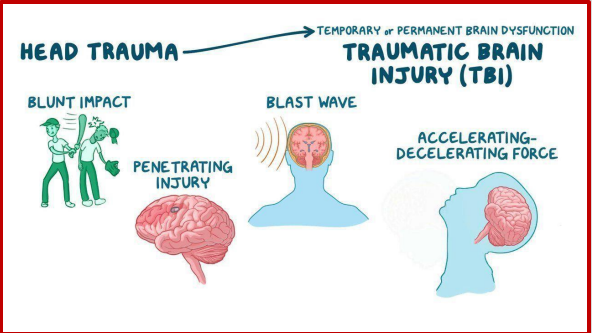
- Riveros Gilardi B, Muñoz López JJ, Hernández Villegas AC, Garay Mora JA, Rico Rodríguez OC, Chávez Appendini R, De la Mora Malvá ez M, Higuera Calleja JA. Types of Cerebral Herniation and Their Imaging Features. Radiographics. 2019 Oct;39(6):1598-1610. doi: 10.1148/rg.2019190018. PMID: 31589570
- Sagittal T1-weighted MR image: Downward displacement of cerebellar tonsils (>5 mm) relative to the dashed line (McRae line)
- Obliteration of the cisterna magna
- Anterior displacement of the medulla (arrow)
- Hydrocephalus (*)

Summary



TIME SINCE ISCHEMIC EVENT	12-24 HOURS	24-72 HOURS	3-5 DAYS	1-2 WEEKS	> 2 WEEKS
Histologic features	Eosinophilic cytoplasm + pyknotic nuclei (red neurons)	Necrosis + neutrophils	Macrophages (microglia)	Reactive gliosis (astrocytes) + vascular proliferation	Glial scar

Histological Changes of ischemic Infarct



THANK YOU!

Lecture and Lab Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>